

Towards comprehensive multi-task land cover change detection leveraging vision-language model and LLM-driven agents

Tengxi Wang^a, Yiru Wang^b, Shuai Zhang^a, Yuxuan Liang^a, Wufan Zhao^a,*

^a The Hong Kong University of Science and Technology (Guangzhou), Guangzhou, 511442, China

^b School of Software Engineering, Beijing Jiaotong University, Beijing, 100044, China

ARTICLE INFO

Keywords:

Change detection
Vision-language model
Multi-agent workflow

ABSTRACT

Land cover change detection is essential for assessing the impacts of urbanization on ecological systems and economic development. As urbanization accelerates, land cover changes involve both semantic transitions and structural modifications, substantially increasing the complexity of detection tasks. However, existing research has primarily focused on 2D change detection, with limited integration of multi-dimensional and multi-modal data. This limitation has restricted the effective modeling of inter-class transitions and spatial structural evolution. Moreover, most methods lack end-to-end pipelines adapted to real-world scenarios, constraining their capacity for comprehensive representation and quantitative impact assessment. To address these challenges, we introduce UrbanChange, the first dataset designed for multi-task (2D and 3D) land cover change detection. It integrates semantic and structural change maps from multi-sensor data with fine-grained change captions, providing a comprehensive foundation for evaluating and analyzing land cover transformations. Leveraging UrbanChange, we develop a vision-language model that directly infers multi-dimensional changes from optical imagery. We further develop GaiaAgent by incorporating this model as a core perception module, forming the agentic workflow for land cover change detection and analysis. GaiaAgent coordinates five task-specific agents in an automated pipeline for instruction interpretation, task decomposition, detection, and quantitative analysis. A Human-in-the-Loop mechanism is incorporated to produce interactive, interpretable reports, enhancing robustness in complex scenarios. Extensive experimental results demonstrate the effectiveness of GaiaAgent. On the UrbanChange dataset, GaiaAgent achieves a 3.9% improvement in F1 and a 0.35 m reduction in cRMSE compared with the strongest baseline. In downstream applications across multiple analytical scenarios, GaiaAgent demonstrates improved reliability and adaptability in land cover change detection and interpretation.

1. Introduction

The rapid acceleration of urbanization has led to complex transformations in land cover, particularly in the areas of urban expansion and ecological degradation (Stilla and Xu, 2023; Liu et al., 2020). These transformations involve not only 2D semantic transitions but also 3D structural modifications, complicating the task of change detection (Marsocci et al., 2023). End-to-end change detection, which accurately identifies areas of transformation and interprets their semantic implications based on land cover categories, has gained significant attention, driven by advances in deep learning technologies (W. Xiao et al., 2024; Kennedy et al., 2009). However, existing studies still predominantly focus on 2D static patterns (Fig. 1(a)), while the interpretation of spatial structures and temporal dynamics remains insufficient for supporting urban renewal analysis.

Optical remote sensing imagery has been shown to implicitly encode 3D structural information through visual representations and contextual information, even without explicit 3D inputs such as Digital Surface Models (DSMs) (Qin et al., 2016; Hong et al., 2026). This capability provides a data-level foundation for integrating 2D semantic changes and 3D structural variations in multi-task land cover change detection. However, most existing methods remain centered on 2D tasks, such as bi-temporal semantic segmentation and change localization (Fig. 1(a) and (b)) (Long et al., 2024; H. Chen et al., 2024; Zhu et al., 2025). Existing multi-task studies are mainly oriented toward building-scale applications, with limited capability to explicitly represent land cover transitions and structural transformation processes (Fig. 1(c)) (Tang et al., 2024; Cui and Jiang, 2023). Notably, the lack of benchmark datasets supporting multi-task land cover change modeling

* Corresponding author.

E-mail address: wufanzhao@hkust-gz.edu.cn (W. Zhao).

<https://doi.org/10.1016/j.isprsjprs.2026.05.025>

Received 5 October 2025; Received in revised form 6 February 2026; Accepted 13 May 2026

Available online 29 May 2026

0924-2716/© 2026 International Society for Photogrammetry and Remote Sensing, Inc. (ISPRS). Published by Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

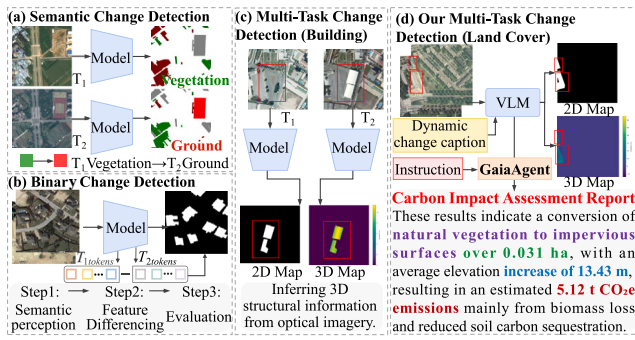


Fig. 1. Summary of change detection task. Vision-language models are applied to multi-task land cover change detection, with the first attempt to employ multi-modal large language model agent for automated workflows.

further constrains systematic analysis of complex urban scenes and structural evolution.

The adoption of foundational and multi-modal models has introduced a novel methodological paradigm in land cover change detection (Dong et al., 2024; Li et al., 2024). These models combine textual semantics with visual features to enable semantic-level change representations, significantly expanding the scope and capabilities of change detection (Zhang et al., 2024). However, despite significant progress in conventional visual tasks, these models continue to face considerable challenges in multi-task land cover change detection, particularly in terms of the insufficient perception of fine-grained land cover characteristics and the precise alignment of text modality with localized change regions (Hao et al., 2024; Y. Chen et al., 2024). These challenges restrict the models' ability to capture fine-grained changes of complex environments.

While representations of change have advanced, the systematic integration of detection and analysis in complex geographical scenarios remains limited. Most existing studies rely on task-specific pipelines. For instance, ChangeMamba and ChangeRD emphasize change detection while neglecting in-depth scene analysis, thereby reducing their generality and flexibility (H. Zhang et al., 2025; Jing et al., 2025). To bridge this gap, incorporating domain knowledge is crucial for interpreting deep learning outputs and transforming them into quantitative impact assessments (Luo et al., 2025). However, the costly acquisition of domain expertise hampers this translation (C. Liu et al., 2024). Recently, intelligent agents powered by multi-modal large language models (MLLMs) have emerged as promising alternatives (Y. Zhang et al., 2025; Marsili et al., 2025). Specifically, ChangeAgent (C. Liu et al., 2024) represents a recent attempt to introduce agent-based reasoning into remote sensing change analysis by enabling instruction-driven interaction. However, its design is primarily oriented toward single-task execution and limited workflow composition, making it difficult to support complex multi-step reasoning and cross-task coordination in urban scenarios. By contrast, multi-agent collaborative paradigms enable dynamic workflow construction aligned with user intent by integrating planning, execution, and interaction into a unified reasoning process (Z. Wang et al., 2025; Y. Xiao et al., 2024). Despite this potential, agent-based frameworks still face persistent challenges, particularly in aligning user instructions with task objectives and ensuring robust reasoning under real-world complexity (Gadot et al., 2025).

To address these challenges, we introduce the first multi-task land cover change detection dataset covering several representative urban areas in the Netherlands. The dataset integrates multi-sensor 2D and 3D change maps with fine-grained change captions, enabling multi-task land cover change detection evaluation. Leveraging this dataset, we develop a vision-language change detection network that jointly captures semantic and structural variations. We further integrate this

Table 1

Comparison of change detection and analysis methods.

Method	3D change detection	Interactive analysis	Complex task reasoning
ChangeMamba (H. Chen et al., 2024)	✗	✗	✗
MTBIT (Marsocci et al., 2023)	✓	✗	✗
Change-Agent (C. Liu et al., 2024)	✗	✓	✗
GaiaAgent	✓	✓	✓

perception model into GaiaAgent (Fig. 1(d)), the multi-agent workflow for change detection and downstream analysis. GaiaAgent decomposes user instructions into structured sub-tasks, coordinates task-specific agents through a modular execution pipeline, and utilizes model outputs to support downstream analytical tasks across diverse application scenarios.

Our contributions are summarized as follows:

- We propose a vision-language model for multi-task land-cover change detection and extend it into a multi-agent workflow, GaiaAgent, for interactive geographical analysis. GaiaAgent follows a structured reasoning paradigm, augmented with an understanding-correction mechanism and human-in-the-loop revision control to enhance task reliability and execution robustness. (Table 1).
- The vision-language change detection model that integrates bi-temporal interaction, semantic-aware positional encoding, and scene-level contrastive learning. Together, these components improve the alignment between change captions and visual representation, while enabling more precise and fine-grained change localization.
- We introduce UrbanChange, the first multi-task land cover change detection dataset. It integrates semantic and structural change maps derived from multi-sensor data with dynamic captions, providing a comprehensive foundation for multi-dimensional land cover change.
- Extensive experiments demonstrate the effectiveness of GaiaAgent. On the UrbanChange dataset, it achieves a 3.9% F1 improvement and a 0.35 m cRMSE reduction over the best baseline. Furthermore, multiple analytical scenarios validate the reliability of GaiaAgent in user-driven tasks, highlighting its practical applicability.

2. Related work

2.1. Remote sensing change detection

Change detection is a fundamental task in remote sensing (Bai et al., 2023; Cheng et al., 2024), typically categorized into Binary Change Detection and Semantic Change Detection (Hu et al., 2023; Ding et al., 2024). Binary change detection (Codegoni et al., 2023) identifies regions of change between bi-temporal images, while semantic change detection (Cui and Jiang, 2023) further classifies land cover categories at each time point through pixel-wise semantic comparisons (Fig. 1(a) and (b)).

Most existing studies focus on 2D change detection, leveraging convolutional neural networks (CNNs) and Transformer based architectures to extract spectral and textural features. However, their semantic reasoning capabilities remain limited in complex urban environments, where ambiguous transitions and diverse land cover patterns pose challenges (Ji et al., 2024). Recent advances in foundation models (e.g., SAM) and vision-language models (e.g., CLIP) have enhanced fine-grained perception and contextual understanding, thereby expanding the scope of 2D change detection (Mei et al., 2024; Gao et al.,

2025). Despite these improvements, 2D methods generally fail to account for vertical structural changes, restricting their effectiveness in capturing the full dynamics of transformation.

To overcome this limitation, 3D change detection methods have been proposed, utilizing elevation information derived from LiDAR, SAR, or DSMs (Qin et al., 2016; A. Xiao et al., 2024; T. Wang et al., 2025), and combining them with semantic segmentation in multi-task frameworks (Marsocci et al., 2023; W. Xiao et al., 2024). Notable methods such as MTBIT and CSCLNet advance feature fusion and multi-scale representation, enabling height inference directly from optical imagery. However, they focus mainly on building-level changes and fall short in systematically modeling diverse land cover transitions. Persistent challenges also remain in integrating semantic and geometric information and in delivering interpretable analyses of change processes.

Based on the above analysis, we propose a unified semantic change detection framework that jointly models both 2D semantics and 3D structural variations for land cover analysis. By directly inferring multi-class land cover transitions and elevation changes from bi-temporal optical imagery, such a framework can improve generalization performance and enhance robustness in real-world scenarios.

2.2. MLLM-driven agentic workflow

Recent advances in MLLMs have significantly enhanced cross-modal semantic alignment, complex task understanding, and tool-augmented reasoning, thereby accelerating the development of multi-agent workflows. By leveraging their capabilities in semantic parsing and inference, agentic workflow interpret high-level tasks and decompose them into modular subtasks, enabling agent collaboration for context-aware execution and adaptive decision-making. This paradigm demonstrates strong potential for constructing scalable and adaptable in remote sensing applications (Liu et al., 2025; Y. Xiao et al., 2024). Existing systems such as M³Builder (Feng et al., 2025) and FilmComposer (Xie et al., 2025) have validated the flexibility of agent-based frameworks in managing complex pipelines across domains such as medical imaging and content generation.

In the field of remote sensing, downstream analysis based on change detection serves as an essential tool for supporting applications such as urban planning and ecological management (W. Xiao et al., 2024; Kennedy et al., 2009). Change-Agent (C. Liu et al., 2024) represents an early attempt to introduce agent-based analysis frameworks into remote sensing change analysis by combining a multi-level change interpretation (MCI) model with an LLM-based agent. The MCI model consists of a pixel-level change detection branch and a semantic-level change captioning branch, with Bi-temporal Iterative Interaction (BI3) layers designed to enhance discriminative feature representation capabilities. However, Change-Agent's workflow remains primarily tailored to executing individual interpretation tasks (e.g., change detection, captioning, object counting, cause analysis) in response to user instructions, without explicit mechanisms for dynamic multi-step reasoning, flexible task composition, or automated coordination across heterogeneous analytical subtasks. As a result, its applicability in supporting integrated high-level geographical reasoning and decision modeling in complex scenarios remains limited.

To further advance change detection toward practical and intelligent applications, we propose a multi-agent workflow tailored for multi-task land cover change detection and interpretation. The framework integrates MLLMs-based agent for task parsing and logical inference, a vision backbone for perceptual grounding, and task-specific agents for execution and feedback-driven refinement. This workflow offers strong flexibility, and scalability, effectively bridging low-level visual perception with high-level semantic decision-making, providing unified support for complex geospatial analysis.

Table 2

Comparison of UrbanChange and other change detection datasets.

Data	N. Images	Dynamic change	Change map	Classes
<i>Binary Change Detection</i>				
LEVIR-CD (Chen and Shi, 2020)	637	✓	2D	1
SYSU-CD (Shi et al., 2021)	20 000	✓	2D	1
CLCD (M. Liu et al., 2022)	600	✓	2D	1
<i>Multi-Task Building Change Detection</i>				
3DCD (Marsocci et al., 2023)	472	✓	2D+3D	1
SMARS (Reyes et al., 2023)	459	✓	2D+3D	1
Hi-BCD (B. Liu et al., 2024)	1500	✓	2D+3D	2
<i>Semantic Change Detection</i>				
HRSCD (Daudt et al., 2019)	582	✗	2D	5
SECOND (Yang et al., 2020)	9324	✗	2D	6
Hi-UCD (Tian et al., 2020)	2586	✗	2D	9
<i>Multi-Task Land Cover Change Detection</i>				
UrbanChange (ours)	1108	✓	2D+3D	6

3. UrbanChange dataset

We propose UrbanChange, the first multi-task dataset designed for semantic-level land cover change detection. As illustrated in Fig. 1(a) and (b) and summarized in Table 2, existing datasets primarily focus on 2D static semantic changes or binary classification tasks, and some are limited to building-level dynamic changes under multi-task settings. Specifically, LEVIR-CD (Chen and Shi, 2020) provide bi-temporal remote sensing imagery to support 2D building change detection. 3DCD (Marsocci et al., 2023) further explore 3D structural changes based on optical imagery and demonstrate their feasibility. In semantic change detection, the HRSCD dataset (Daudt et al., 2019) delineates change regions from bi-temporal imagery and conducts independent semantic segmentation at each time point, enabling the characterization of category-level transitions between T_1 to T_2 .

To enable more comprehensive analysis of land cover dynamics, it is important to integrate 2D semantics, 3D structures, and temporal transitions for multi-dimensional semantic change detection. UrbanChange mitigates the limitations of existing datasets by integrating fine-grained change captions, dynamic semantic annotations, and 3D structural representations. This comprehensive design enhances the capacity for multi-dimensional semantic change analysis and supports a range of downstream applications, including environmental assessment and urbanization monitoring.

3.1. 2D and 3D image modalities

We collected representative regional data from four areas—Amsterdam, Rotterdam, The Hague, and West Friesland—based on the Beeldmateriaal Nederland (BN)¹ and Actueel Hoogtebestand Nederland (AHN)² platforms, covering diverse environments ranging from rural to urban and from traditional regions to highly modernized areas (as shown in Fig. 2(a)). The DSMs data were acquired via airborne LiDAR scanning by extracting the highest surface points and generating interpolated surfaces. The optical imagery was obtained through aerial photography and processed with geometric correction and orthorectification to produce high-precision orthophotos. Each city tile measures 20,000 × 25,000 pixels, with a spatial resolution of 0.25 m. After preprocessing and spatial alignment, a total of 1108 valid samples were obtained and randomly divided into training (70%), validation (11%), and testing (19%) sets, ensuring no data leakage from the training set.

The image modality in UrbanChange is constructed through manual annotation, with the structure and overview of change categories illustrated in Fig. 2(b). The six pre-defined change types capture representative land cover transitions, including urban expansion, ecological

¹ <https://www.beeldmateriaal.nl/>

² <https://www.ahn.nl/>

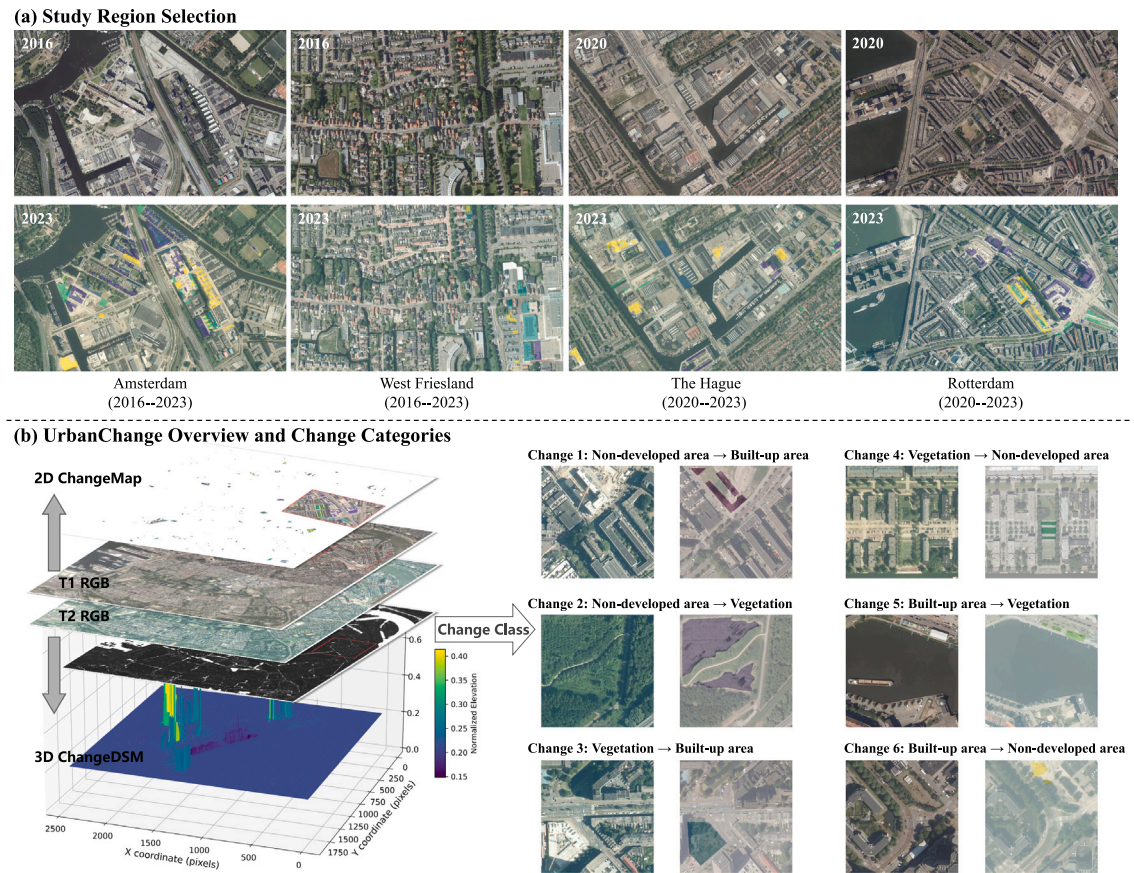


Fig. 2. UrbanChange dataset. It covers four representative urban regions in the Netherlands: Amsterdam (2016–2023), West Friesland (2016–2023), The Hague (2020–2023), and Rotterdam (2020–2023). The dataset provides 2D change maps, 3D change maps, and dynamic captions for each change category.

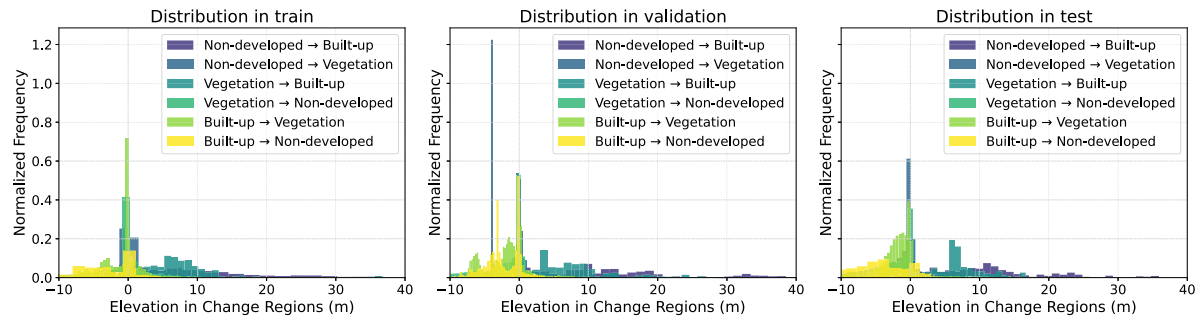


Fig. 3. Visualization of the change height distribution for each category.

degradation, and restoration, and are characterized by clear semantic separability and practical applicability in real-world scenarios. The data construction process involves the following steps: (1) Using manually annotated change maps to mask the DSMs data and extract height information from bi-temporal change regions; (2) Computing the height difference between DSMs pairs and converting anomalous noise values to zero, thereby generating ground-truth height labels reflecting real-world structural variation; (3) Cropping both imagery and labels into 400×400 pixel patches, while removing non-change samples to ensure the effectiveness and representativeness of the data distribution.

We further conducted statistical analysis of both 2D and 3D change information in the UrbanChange dataset. The proportion of change areas varies across the four study regions, with Amsterdam exhibiting the highest change ratio (1.58%) and The Hague the lowest (0.70%), as detailed in Table 3. Moreover, to characterize the spatial structural

Change categories	Change ratio (%)				Changed objects
	Amsterdam	West Friesland	The Hague	Rotterdam	
Non-developed → Built-up	1.19%	0.49%	0.73%	1.15%	507
Non-developed → Vegetation	0.79%	0.07%	0.18%	0.30%	205
Vegetation → Built-up	0.28%	1.35%	0.10%	0.34%	334
Vegetation → Non-developed	0.29%	0.33%	0.52%	0.67%	392
Built-up → Vegetation	0.10%	0.45%	0.03%	0.02%	140
Built-up → Non-developed	0.30%	0.14%	0.35%	0.68%	254
Total Change Ratio	2.95%	2.83%	1.92%	3.17%	–
Samples w/ change	416	226	283	183	–

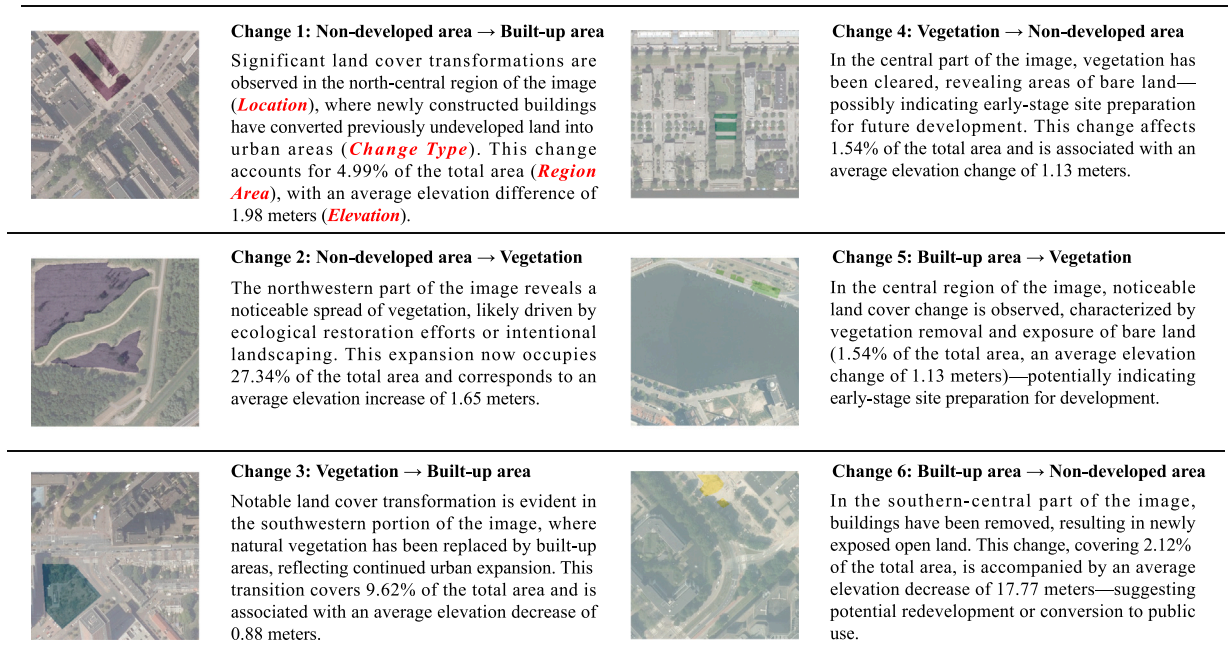


Fig. 4. Visualization of dynamic captions in UrbanChange. The captions specifically include change locations, change types, area of change, and average change elevation.

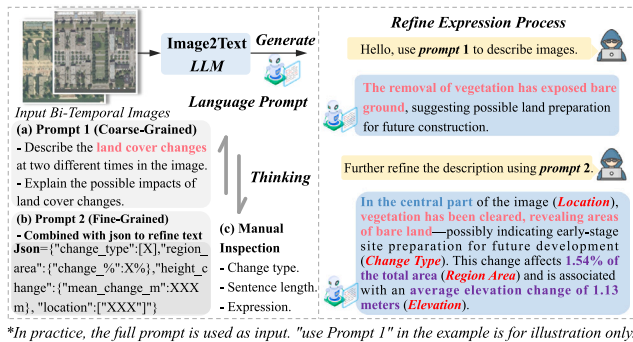


Fig. 5. Workflow of fine-grained change captions generated by a large language model through a multi-step process.

information of different changes, we visualized the elevation distribution of 3D changes (Fig. 3). The 3D change patterns capture variations in vertical structure associated with surface cover transitions, providing complementary information to 2D semantics.

3.2. Dynamic textual modality

Existing multi-modal studies have shown that high-quality textual information enhances representation learning by embedding semantic context into visual representations (Hao et al., 2024). Therefore, we employ the large language model (LLM) GPT-4o to automatically generate dynamic captions for change scenes, which describe change processes between bi-temporal remote sensing images rather than static single-time scenes. To jointly capture global change context and fine-grained visual details, a two-stage captioning strategy is adopted (examples are shown in Fig. 4, the pipeline in Fig. 5, and prompt templates in Appendix B). The captioning process involves the following steps: (1) The bi-temporal images and a global descriptive prompt are first fed into the LLM to generate an initial change captions aligned with the input imagery, serving as a semantic scaffold for further refinement. (2) Based on the 2D and 3D change maps, a structured JSON file is

generated containing change-related information such as location, type, area, and average elevation. This is then used as an input prompt for the LLM to obtain more detailed and accurate descriptions. This process enhances the spatial granularity of information and enables automatic correction of inconsistencies between the change captions and the input images. (3) To ensure accuracy and reduce hallucinations produced by the LLM, the final outputs are reviewed by postgraduate annotators with backgrounds in remote sensing and computer science to verify semantic and factual correctness.

4. Methodology

We propose a vision-language model for multi-task land cover change detection and further develop a multi-agent workflow built upon it to enable automated detection and analysis in complex geographic scenarios, as shown in Fig. 6. Specifically, the vision-language model jointly perceives image and textual information to enhance the comprehensive representation of land cover changes, while the multi-agent system collaborates to decompose complex tasks into chain-of-thought subtasks and generate analytical prompts via the vision-language model, enabling in-depth interpretation of geographic changes.

4.1. Vision-language change detection model

To enable effective and accurate change detection, we design a multi-modal network guided by change-aware captions (Fig. 7). During training, the network is trained with bi-temporal remote sensing images and fine-grained change captions as inputs, under the supervision of 2D and 3D change maps, while the captions are embedded into the visual branch to enhance change-aware perception. In the inference stage, the network relies solely on the visual branch to process images for multi-task change detection.

Visual Representation Learning. To enhance sensitivity to fine-grained changes and establish a foundation for cross-modal representation learning, we integrate the information interaction module and semantic encoding module in parallel to model bi-temporal images.

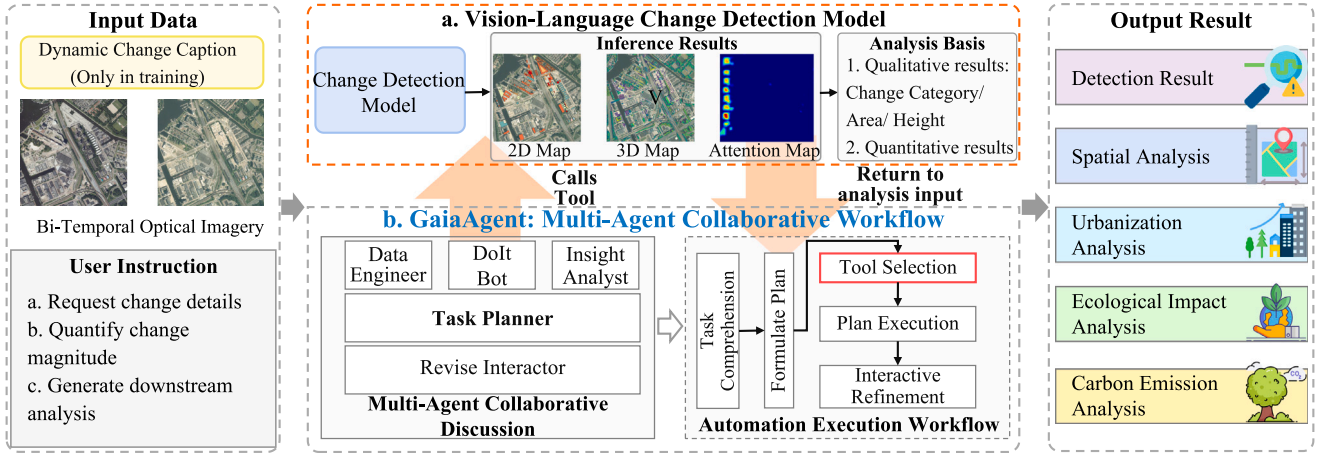


Fig. 6. Overall architecture of the proposed GaiaAgent framework for downstream analysis, illustrating the integration of a vision-language change detection model with multi-agent collaborative analysis. The change detection network provides 2D and 3D change maps from bi-temporal imagery as detection results, upon which the agent workflow performs task-driven and interactive downstream analysis.

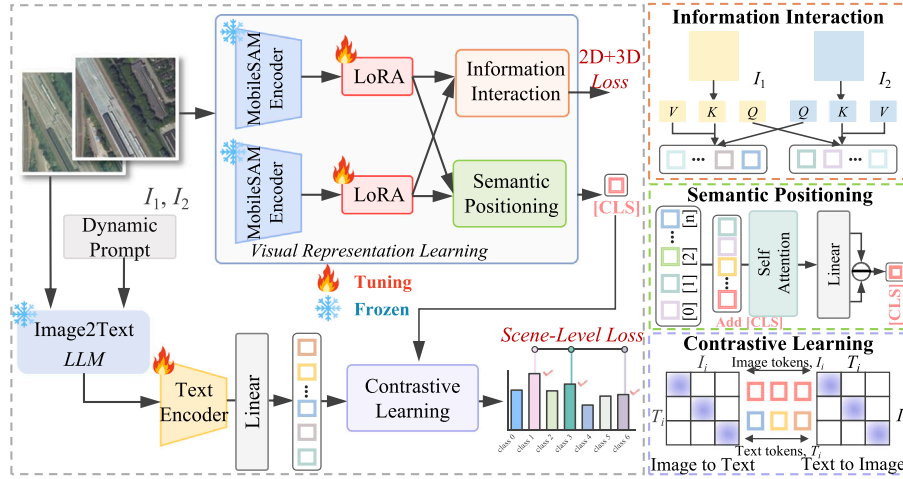


Fig. 7. Architecture of the vision-language model for change detection. The network is trained using bi-temporal images and change captions under 2D&3D supervision, while inference relies solely on bi-temporal images for multi-task change detection.

We adopt MobileSAM to extract semantic features from I_1 and I_2 , leveraging its strong perception of spatial structures (Zhang et al., 2023). We incorporate Low-Rank Adaptation (LoRA) (Hu et al., 2021) into the encoder attention modules for parameter-efficient transfer of semantic representations. This strategy alleviates the domain shift between pre-training data and remote-sensing imagery and enhances semantic stability.

$$F_i = \text{Encoder}(I_i; W + \Delta W) = \text{Encoder}(I_i; W + AB) \quad (1)$$

where W denotes the frozen pre-trained weight matrices in the MobileSAM encoder, $A \in \mathbb{R}^{d \times r}$, $B \in \mathbb{R}^{r \times d}$ are the low-rank matrices introduced by LoRA (with rank $r \ll d$), and F_i is the resulting visual features.

In the representation stage, we introduce an information interaction module to explicitly model cross-temporal interactions between F_1 and F_2 , enhancing the representation of change patterns. Specifically, the attention mechanism assigns higher weights to real change regions and suppresses pseudo changes caused by noise and illumination variations. The bi-temporal features F_1 and F_2 are projected into Query Q_i , Key K_i , and Value V_i representations, where Q_1 aggregates the key-value pairs (K_2, V_2) from F_2 and Q_2 symmetrically attends to F_1 , enabling bi-directional information exchange. This stage is defined as:

$$\text{Attn}_1 = \text{Softmax}\left(\frac{Q_1 K_2^T}{\sqrt{d_k}}\right) V_2, \text{Attn}_2 = \text{Softmax}\left(\frac{Q_2 K_1^T}{\sqrt{d_k}}\right) V_1 \quad (2)$$

The outputs Attn_i explicitly model cross-temporal feature correlations and capture fine-grained temporal differences through bi-directional interaction. To further characterize change information, we apply element-wise subtraction to the bi-temporal features to obtain a difference sequence, which explicitly captures temporal change patterns for subsequent supervised learning.

In parallel with the information interaction module, we introduce a semantic encoding module that enhances patch-level features via semantic-guided positional encoding and aggregates global semantics to produce discriminative embeddings for subsequent learning. Specifically, feature F_i is first flattened into a token sequence. Motivated by the ViT paradigm (Dosovitskiy, 2020) that leverages learnable positional embeddings to capture spatial layouts, we further incorporate a semantic-conditioned linear projection to guide positional encoding, enabling spatial representations to dynamically adapt to semantic variations, which is formulated as:

$$PE_i = P_i + F_i W^P + b^P, \quad i \in \{1, 2\} \quad (3)$$

where P_i denotes the base learnable positional embedding, and W^P and b^P are learnable parameters of the linear projection layer. Through this positional encoding, semantic information is injected into spatial representations, enabling adaptive spatial bias adjustment and improved modeling of spatial structures. A learnable [CLS] token is then

concatenated with the token sequence and interacts with all visual tokens through self-attention to aggregate global semantics. Finally, the [CLS] token extracted from the change-aware difference features is projected to a fixed 512-dimensional space as the visual embedding a_I^G for cross-modal alignment with textual representations.

Dynamic Textual Representation Learning. Existing studies have demonstrated that accurate textual captions can effectively enhance visual perception in vision-language models (Yan et al., 2024b). Therefore, we leverage textual embeddings to guide the model's attention toward change regions, improving its understanding of spatial structures. Specifically, a prompt is designed to guide GPT-4o in generating fine-grained change captions D . To capture internal dependencies and align text with visual changes, token embeddings are processed through attention mechanism.

Cross-Modal Representation Enhancement. We propose a scene-level contrastive learning strategy to align change semantics between images and text, enhancing textual guidance for visual representation learning. Notably, change captions provide a scene-level semantic summary by integrating one or multiple localized change events, rather than offering region-wise one-to-one correspondence annotations. This naturally motivates the adoption of scene-level semantic alignment as the cross-modal learning objective.

Specifically, in the visual branch, we adopt a 512-dimensional change-aware embedding a_I^G , which aggregates semantic-guided patch features via self-attention to represent scene-level change semantic context. In the textual branch, we adopt the final feature a_T from the CLIP text encoder, which aggregates semantic information from one or multiple change events in the captions (e.g., change types, spatial distributions, and change magnitudes). a_T is then projected into a 512-dimensional representation space via a linear layer, formulated as $a_T = \text{Linear}(T)$. Based on a_I^G and a_T , we introduce a symmetric contrastive loss $\mathcal{L}_{\text{Con}}$ to promote scene-level semantic alignment between visual and textual representations, consisting of the image-to-text term $\mathcal{L}_{\text{Image} \rightarrow \text{Text}}$ and the text-to-image term $\mathcal{L}_{\text{Text} \rightarrow \text{Image}}$:

$$\mathcal{L}_{\text{Con}} = -\log \frac{\exp(\text{sim}(a_I^G, a_T^L)/0.1)}{\sum_{i=1}^n \exp(\text{sim}(a_I^G, a_{T,i}^L)/0.1)} - \log \frac{\exp(\text{sim}(a_T^L, a_I^G)/0.1)}{\sum_{i=1}^n \exp(\text{sim}(a_T^L, a_{I,i}^G)/0.1)} \quad (4)$$

where $\text{sim}(\cdot)$ is the similarity function. τ is the temperature parameter and is empirically set to 0.1 in our experiments. The contrastive loss $\mathcal{L}_{\text{Con}}$ encourages semantic alignment between scene-level textual embeddings and scene-level visual representations in a shared embedding space.

Multi-task Change Detection Loss Function. During training, the 2D and 3D branches are supervised by Focal Loss and mean squared error (MSE) loss, respectively, and jointly optimized via a weighted summation. In addition, $\mathcal{L}_{\text{Con}}$ is used to explicitly align visual features with fine-grained textual semantic embeddings, leading to more accurate change-aware visual representations.

$$\mathcal{L}_{2D} = -\frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W \alpha_i \left(1 - p_i^{(i,j)}\right)^\gamma \log(p_i^{(i,j)}) \quad (5)$$

$$\mathcal{L}_{3D} = \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W \left(H_{i,j} - \hat{H}_{i,j}\right)^2 \quad (6)$$

$$\mathcal{L} = 2 \cdot \mathcal{L}_{2D} + 1 \cdot \mathcal{L}_{3D} + 0.1 \cdot \mathcal{L}_{\text{Con}} \quad (7)$$

where H and W denote the height and width of the prediction map. For each pixel (i, j) , $p_i^{(i,j)}$ is the predicted softmax probability of the ground-truth class, α_i is the class-specific weighting factor, and γ is the focusing parameter of the focal loss. For the 3D task, $H_{i,j}$ and $\hat{H}_{i,j}$ denote the ground-truth and predicted height values, respectively. $\mathcal{L}_{\text{Con}}$ weighted by 0.1 is included in the overall objective to incorporate fine-grained textual semantics into the visual branch.

4.2. Automated agentic workflow

The long-chain reasoning paradigm (Ebrahim et al., 2025) emphasizes decomposing complex problems into executable subtasks and performing multi-step reasoning through intermediate state propagation and tool invocation. Based on this principle, we construct a stage-wise analytical workflow and implement multi-agent collaboration on the Lagent library.³ GaiaAgent (Fig. 8) consists of five functional modules: Task Planner, Data Engineer, DoIt Bot, Insight Analyst, and Revision Interactor. The DoIt Bot integrates a vision-language model for change detection and quantitative output generation. These agents collaborate through information sharing and task allocation to enable automated detection and analysis of complex scenarios.

- **The Task Planner.** It employs chain-of-thought reasoning to interpret and decompose user instructions related to change detection, generating a detailed execution plan that specifies the required operations and tool invocations for each stage (Prompt as shown in the Appendix A.1).
- **The Data Engineer.** It inspects and refines the user-provided data to ensure its format is suitable for change detection and subsequent processing (A.2).
- **The DoIt Bot.** It executes the planned workflow by invoking appropriate tools, performing change detection computations on the preprocessed data, and generating analysis basis for downstream tasks (A.3).
- **The Insight Analyst.** It integrates quantitative indicators with qualitative results to analyze the detected changes, producing outputs in both structured formats and natural language explanations (A.4).
- **The Revise Interactor.** It interacts with the human user and applies stylistic adjustments or initiates extended analyses based on user input, enabling the generation of analysis reports under a human-in-the-loop framework (A.5).

The overall workflow is divided into two main stages:

Contextual Task Interpretation. GaiaAgent performs task interpretation and plan formulation during the Contextual Task Interpretation stage, where the Planner autonomously interprets user instructions and collaborates with other agents to generate the execution plan.

The instruction understanding process simulates an implicit dialog: given an instruction I and input image X , the Planner extracts task goals G (e.g., “detect changes”) and constraints C (e.g., tool usage), which form the semantic foundation of the task. For complex cases involving visual input or quantitative analysis, the Planner applies its reasoning capability ϕ to refine goals and constraints into a fine-grained task list T , which directly forms the basis of the initial executable plan P_0 . To ensure semantic alignment, the Planner applies an Understanding-Correction mechanism by computing an alignment score $\alpha = \psi(I, P_0)$. If $\alpha < \delta$, a self-correction loop is triggered to refine the G and C , adjust T , and produce an updated plan P that ensures more reliable downstream execution.

Guided by chain-of-thought reasoning strategy that evaluates task analyzability, data availability, and data usability, the Planner collaborates with the DoIt Bot, Insight Analyst, and Data Engineer to finalize the execution plan. As shown in Algorithm 1, GaiaAgent leverages structured communication and plan-level coordination to execute tasks precisely while mitigating semantic drift. Specifically, the Planner parses P to extract a task set $A = \{a_1, a_2, \dots, a_n\}$, and first consults the Insight Analyst to verify goal-method alignment for each subtask by confirming whether the requested outputs are semantically analyzable, producing guidance Q_i . The Planner then coordinates with the DoIt Bot to generate a task-specific configuration θ_i and assess feasibility under

³ <https://github.com/InternLM/lagent>

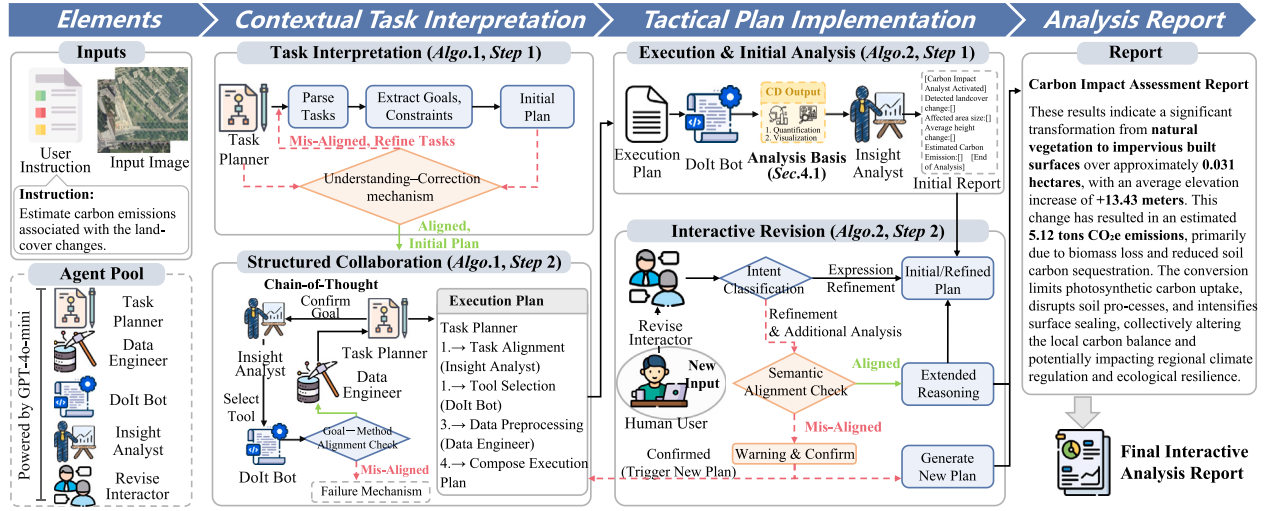


Fig. 8. Agentic workflow of GaiaAgent. The workflow consists of two stages: Stage 1 interprets user instructions and generates an execution plan; Stage 2 executes the plan via collaborative agents, supported by a vision-language model.

the available tool (Vision-Language Change Detection Model) and data constraints, where θ_i determines the required 2D/3D inputs. If a task is identified as semantically non-analyzable (e.g., due to goal-method mismatch), GaiaAgent explicitly returns a task cannot be analyzed state instead of forcing execution. To ensure compatibility with the DoIt Bot, the Data Engineer preprocesses each input image x_i to produce x'_i through validation and necessary resizing or patch partitioning. After preprocessing, the Planner assembles executable units into the execution plan $EP \leftarrow \{(x'_i, T(\theta_i), Q_i)\}$, where $T(\theta_i)$ denotes the invocation of the change detection network under the task-specific configuration θ_i .

Algorithm 1: Structured Plan Execution and Agent Communication

```

Input: Instruction  $I$ , input image  $X$ 
Output: Execution plan  $EP$ 
/* Step 1: Task Interpretation */
1  $G, C \leftarrow \text{Planner.parse}(I, X)$ ;
2  $P_0 \leftarrow T \leftarrow \phi(G, C)$ ; // Task list
3  $\alpha \leftarrow \psi(I, P_0)$ ; // Understanding-Correction
4 while  $\alpha < \delta$  do
5    $P \leftarrow \text{Planner.refine\_plan}(I, P_0)$ ;
6    $\alpha \leftarrow \psi(I, P)$ ;
/* Step 2: Agent communication for plan action */
7  $A = \{a_1, a_2, \dots, a_n\} \leftarrow \text{Planner.extract\_tasks}(P)$ ;
8 foreach  $a_i \in A$  do
9    $Q_i \leftarrow \text{InsightAnalyst.confirm\_goal}(a_i.\text{required\_output})$ ;
   // Goal-output alignment
10   $\theta_i \leftarrow \text{DoItBot.configure}(Q_i)$ ; // 2D/3D inputs
11  if not analyzable then
12    Return "Task cannot be analyzed"
13   $x'_i \leftarrow \text{DataEngineer.preprocess}(x_i)$ ;
14  Append  $(x'_i, T(\theta_i), Q_i)$  to  $EP$ ; // Invoke network  $T$  with config  $\theta_i$ 
15 return  $EP$ 

```

Tactical Plan Implementation. In the tactical plan implementation stage, GaiaAgent generates analytical results and enables interactive refinement of outputs. Guided by the predefined execution plan EP , the DoIt Bot and Insight Analyst process the input imagery to produce a structured report (Fig. 8). To improve interpretability and flexibility, GaiaAgent introduces a Revise Interactor that enables dynamic report

customization and optional re-analysis without reactivating the full workflow.

Algorithm 2: Execution and Analysis with Interactive Revision

```

Input: Execution plan  $EP$ , input image  $X$ 
Output: Final structured analysis report  $R$ 
/* Step 1: Execution and analytical assessment */
1  $\{q_i, v_i\} \leftarrow \text{DoIt Bot}(EP, x_i)$ ,  $R \leftarrow \text{Analyst}(EP, q_i, v_i)$ ;
/* Step 2: Interactive Revision */
2 if user requests revision then
3    $s \leftarrow \phi(\text{user\_new input})$ ; // Intent classification
4   if  $s \in \{\text{academic, conversational, visual}\}$  then
5      $r' \leftarrow \psi(R, s)$ ; // Style-specific rewriting
6      $R \leftarrow r'$ 
7   else if  $s \Rightarrow \text{new analysis then}$ 
8      $\alpha \leftarrow \psi(R, s)$  if  $\alpha < \delta$  then
9        $R \leftarrow \text{Analyst}'(R, s)$ ; // Extended reasoning
10    else
11       $\omega \leftarrow \text{confirm}()$ ; // Warning
12      if  $\omega = \text{True}$  then
13         $P' \leftarrow \text{Planner}(s)$ ;
14         $EP' \leftarrow \text{DoIt Bot}(P')$ ;
15         $R \leftarrow \text{Analyst}(EP')$ 
16 return  $R$ 

```

As shown in Algorithm 2, a human-in-the-loop mechanism is employed following the generation of the initial report. First, the DoIt Bot produces intermediate results $\{q_i, v_i\}$, which are subsequently interpreted by the Insight Analyst into a structured report R . Next, the Revise Interactor categorizes user requests s (e.g., stylistic edits, visualization updates, or new analytical queries). Stylistic edits are applied directly to R , whereas analytical requests are assessed for semantic alignment with the existing report. If alignment is confirmed, the task is delegated to the specific agent; otherwise, the user is prompted for confirmation. This human-in-the-loop validation serves as a safeguard against semantic drift and hallucinated content. Upon confirmation, the Planner generates a revised plan P' , the DoIt Bot executes it to produce updated results EP' , and the Insight Analyst correspondingly updates R .

Table 4
Results of comparison on the UrbanChange dataset.

Method	Year	2D change detection		3D change detection	
		F1 (%) ↑	mIoU (%) ↑	RMSE (m) ↓	cRMSE (m) ↓
2D Change Detection Methods					
FC-EF (Daudt et al., 2018)	2018	33.41	25.47	3.08	14.53
ChangeFormer (Bandara and Patel, 2022)	2022	31.35	24.16	3.57	15.39
ChangeAgent (C. Liu et al., 2024)	2024	46.54	34.05	2.63	12.69
ChangeMamba (H. Chen et al., 2024)	2024	<u>53.27</u>	<u>39.49</u>	2.29	10.77
ChangeRD (Jing et al., 2025)	2025	45.95	34.52	2.47	12.35
Multi-Task Change Detection Methods					
Siamese ResU-Net (Marsocci et al., 2023)	2023	42.48	30.96	2.91	13.41
MTBIT (Marsocci et al., 2023)	2023	50.73	37.51	2.02	<u>10.63</u>
CSCLNet (W. Xiao et al., 2024)	2024	52.01	38.77	<u>1.96</u>	10.71
GaiaAgent	2025	57.17	43.64	1.95	10.28

5. Experiments

5.1. Experimental setup

Implementation Details. For multi-agent collaboration, we adopt OpenAI's GPT-4o-mini model to simulate different roles and enable cooperative functionality. For the vision-language model, training is conducted on an NVIDIA RTX 4090 GPU with 24 GB memory, using a batch size of 2 for a total of 200 epochs. The Adam optimizer is employed with a learning rate of 1×10^{-4} , and the random seed is set to 18. The visual encoder uses the TinyViT from MobileSAM, with LoRA settings of rank = 16 and $\alpha = 32$. The text encoder is based on the CLIP ViT-B/32. For $\mathcal{L}_{2D}$, the class weights α_i are set according to the class distribution as [0.3, 2.0, 2.0, 3.0, 1.5, 0.8, 1.0].

Scenario Setup for Downstream Analysis. We design the multi-agent analytical scenarios to support land cover-induced carbon emission assessment (Ou et al., 2024; Qiu et al., 2024), ecological analysis (Yan et al., 2024a), and spatial statistics (Xiao et al., 2025). The detailed computational formulations are summarized in Table 11, where the emission factors for different land cover types are adopted from Wang et al. (2019). The emission factor values used in this study are $EF_{veg} = 13.27$, $EF_{bare} = 5.65$, and $EF_{build} = 29.80$ kg/m².

Evaluation Metrics. We adopt commonly used evaluation metrics to assess the performance of the model. For 2D tasks, we evaluated the model's accuracy using mean Intersection over Union (mIoU) and F1. For 3D tasks, Root Mean Square Error (RMSE) and change-specific RMSE (cRMSE) were employed (Marsocci et al., 2023). Specially, the cRMSE metric focuses exclusively on errors within changed areas, providing a more precise assessment of the model's 3D perception capability in those regions.

$$cRMSE = \sqrt{\frac{1}{n_c} \sum_{i=1}^{n_c} (\widehat{\Delta H}_i^C - \Delta H_i^C)^2} \quad (8)$$

where, $\widehat{\Delta H}_i^C$ represents the predicted pixel height in the change region, and ΔH_i^C represents the ground truth pixel height in the change region.

Baselines. We compare GaiaAgent with several representative remote sensing change detection methods. Specifically, ChangeAgent (C. Liu et al., 2024) combines pixel-level detection with LLM-based semantic interpretation; ChangeMamba (H. Chen et al., 2024) utilizes a spatiotemporal state space model to capture global dependencies; ChangeRD (Jing et al., 2025) improves spatial consistency through registration-based refinement; MTBIT and Siamese ResUNet (SUNet) (Marsocci et al., 2023) learn semantic features and image differences to enhance spatial consistency and detection accuracy; and CSCLNet (W.

Xiao et al., 2024) enhances multi-scale spatiotemporal features via cross-temporal attention. Additionally, we include classical methods such as FC-EF (Daudt et al., 2018) and ChangeFormer (Bandara and Patel, 2022) as baselines. To ensure fairness in comparison, for methods lacking 3D prediction capability, we incorporate GaiaAgent's 3D module into their architectures, enabling consistent evaluation across both 2D and 3D dimensions.

5.2. Quantitative analysis

Comparative experiments with state-of-the-art (SOTA) methods validate the effectiveness of GaiaAgent in both 2D and 3D change detection tasks. The evaluation results on the UrbanChange dataset are presented in Table 4. In the 2D change detection task, GaiaAgent achieves the highest F1 of 57.17%, outperforming the strongest baseline, ChangeMamba, by 3.9%. Compared to early methods such as FC-EF and ChangeFormer, the gain exceeds 20%, demonstrating significant improvement in semantic change recognition. GaiaAgent also yields the highest mIoU of 43.64%, indicating superior semantic consistency in complex urban scenarios. Notably, even compared with advanced multi-task baselines like MTBIT (37.51%) and CSCLNet (38.77%), GaiaAgent maintains a clear advantage, validating its ability to jointly leverage semantic and geometric information.

In the 3D domain, GaiaAgent demonstrates consistent superiority over competing methods. It achieves the lowest RMSE (1.95 m) and a competitive cRMSE (10.28 m), outperforming both MTBIT and CSCLNet. Notably, GaiaAgent shows greater robustness in cRMSE, indicating more stable height estimation behavior and improved geometric consistency across diverse land cover types. Such performance advantages enable more reliable characterization of fine-grained 3D deformations, which is particularly relevant for elevation-aware applications, including terrain change monitoring and volumetric change estimation.

5.3. Qualitative analysis

Detection Qualitative Analysis. To intuitively showcase the effectiveness of change detection, we selected several well-performing models—GaiaAgent, CSCLNet, and ChangeMamba—for visual comparison, as illustrated in Fig. 9. It is clear that GaiaAgent outperforms the other models in terms of change region localization, semantic interpretation, and spatial structure perception. In complex scenarios, many models struggle with mis-classification due to the diversity of change categories and the similarity of semantic features in the changed regions. However, GaiaAgent benefits from the integration of textual semantic understanding during training. This enhancement allows GaiaAgent to better interpret the underlying meaning of changes, enabling it to focus on the correct regions. As a result, GaiaAgent consistently demonstrates

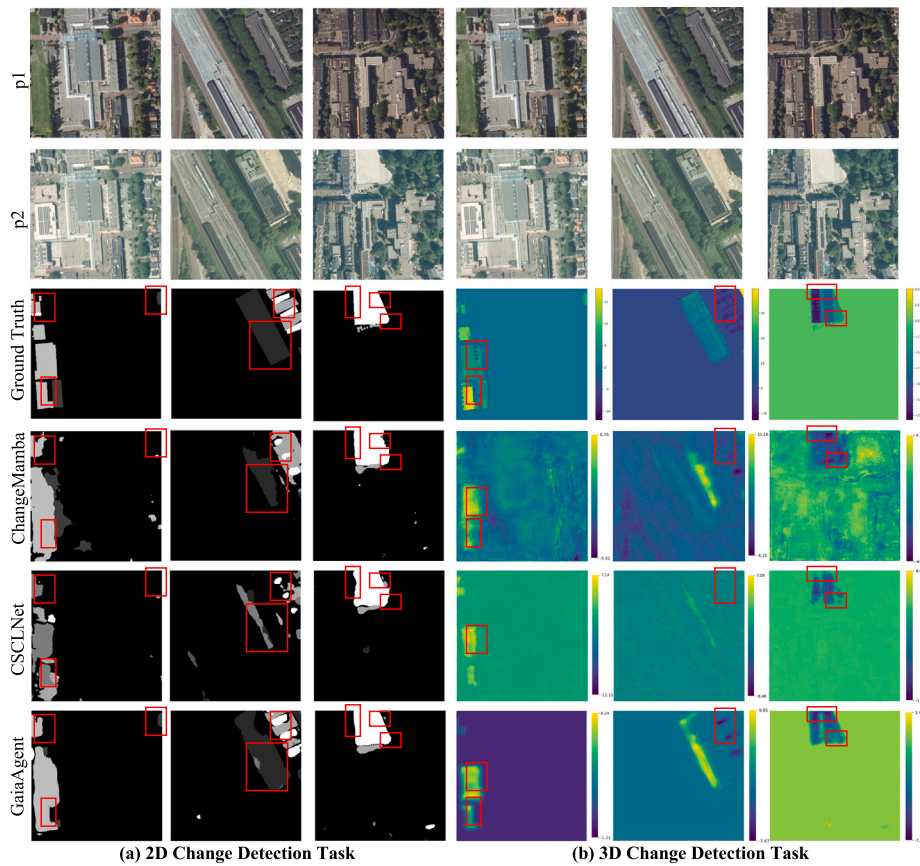


Fig. 9. Qualitative comparison of test results on the UrbanChange dataset, showing (a) 2D task outputs and (b) 3D task outputs, with color bars indicating height in meters. ChangeMamba and CSCLNet are included for visual comparison with GaiaAgent.

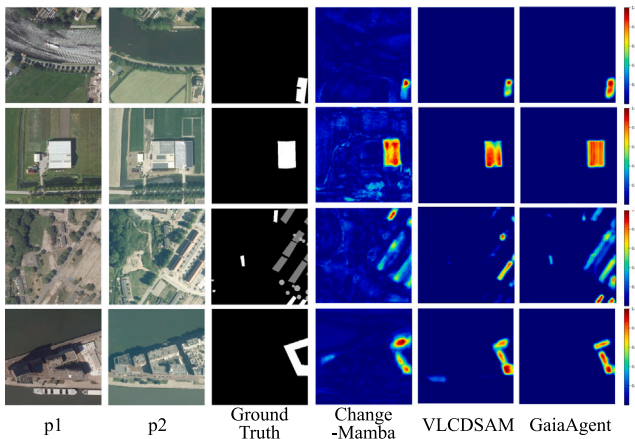


Fig. 10. Fine-grained text-guided semantic attention maps for change detection. GaiaAgent shows improved focus and clarity compared to ChangeMamba and VLCD SAM.

superior performance, accurately capturing changes with greater clarity and precision.

As shown in Fig. 10, to further investigate the impact of textual guidance, we visualize attention maps over bi-temporal change regions. The results clearly show that the visual network guided by dynamic text can effectively focus attention on the correct change areas compared to both the VLCD SAM (GaiaAgent w/o textual modality) and ChangeMamba. Combined with the evaluation results of GaiaAgent, it suggests

that the effectiveness of land cover change detection is directly related to the network's ability to allocate and focus perception accurately.

Interaction Qualitative Analysis. Fig. 11 illustrates the interactive workflow between GaiaAgent (multi-agent framework) and the Change-Agent(single-agent baseline), where the Change-Agent independently performs task understanding and prompt construction for each application scenario to support downstream analysis. The revision interaction mechanism enables GaiaAgent to incorporate user feedback, correctly identify task types, and perform constrained analysis within the available data scope, thereby reducing hallucination risks in LLM reasoning.

Specifically, Figs. 11(a) and (b) illustrate that certain analytical queries exceed the reasoning scope of the Change-Agent when revision is not applied, leading to outputs that lack sufficient evidential support. With the revision mechanism enabled, these unsupported requests are instead filtered and returned with bounded responses, avoiding over-interpretation and unreliable inference. For feasible analytical tasks, the revision stage exhibits two behaviors. As shown in Figs. 11(c) and (d), previously generated analytical outputs are reused to support follow-up reasoning, enabling consistent carbon emission and ecological degradation assessment. Meanwhile, Figs. 11(e) and (f) demonstrate that the revision mechanism can further refine report presentation by improving clarity and structural organization, while preserving the quantitative results.

5.4. Ablation experiments

To further evaluate the effectiveness of each module in GaiaAgent, we perform ablation studies on key components, as shown in Table 5.



Fig. 11. Visualization of interactive revision for downstream analytical tasks. This figure compares analysis results under non-interactive and revision-enabled settings across multiple downstream tasks.

Table 5
Ablation study of GaiaAgent on 2D and 3D change detection tasks.

Module	2D change detection		3D change detection	
	F1 (%)	mIoU (%)	RMSE (m)	cRMSE (m)
GaiaAgent w/o textual modality	54.62	40.88	2.03	10.66
GaiaAgent w/o Prompt2 refinement	54.81	40.93	1.98	10.29
GaiaAgent w/o Info-Inter	55.12	41.59	2.03	10.36
GaiaAgent w/o Sem-Enc	55.09	41.80	1.95	10.68
GaiaAgent	57.17	43.64	1.95	10.28

Effects of Textual Modality. We compare GaiaAgent with VLCD SAM (GaiaAgent w/o textual modality), which retains only the visual encoder, to assess the impact of the textual modality on performance. Experimental results demonstrate that incorporating the textual modality consistently improves the performance of GaiaAgent in both 2D

and 3D change detection. Specifically, the F1 in the 2D task increases by 2.55%, while the cRMSE in the 3D task is reduced by 0.38 m. Combined with Fig. 10, the textual modality improves the network's focus on change regions, guiding the model to concentrate on change areas rather than background or irrelevant regions. Throughout this process, the textual modality introduces explicit semantic constraints and change priors during training, which are effectively embedded into the learning process and enable the visual encoder to learn more discriminative spatial representations.

Effects of Fine-Grained Change Captions. To further investigate the contribution of fine-grained change caption to the performance of GaiaAgent, we conduct an ablation study using only the coarse-grained captions generated by Prompt1 (GaiaAgent w/o Prompt2 refinement). Under this setting, the fine-grained captions produced by the proposed two-stage generation strategy consistently outperform the coarse-grained captions when integrated into GaiaAgent. Specifically,

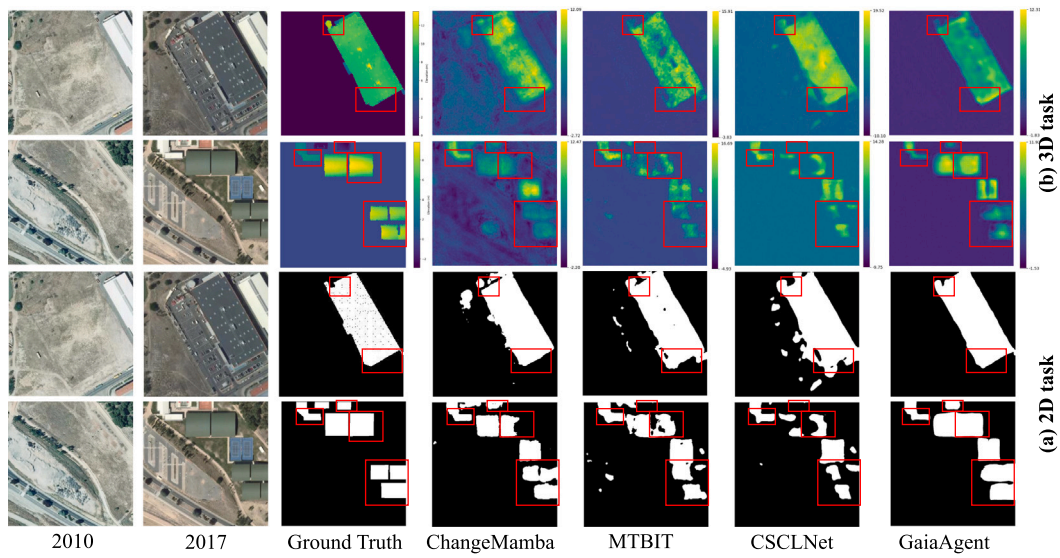


Fig. 12. Qualitative comparison of test results on the 3DCD dataset. ChangeMamba, MTBIT and CSCLNet are included for visual comparison with GaiaAgent.

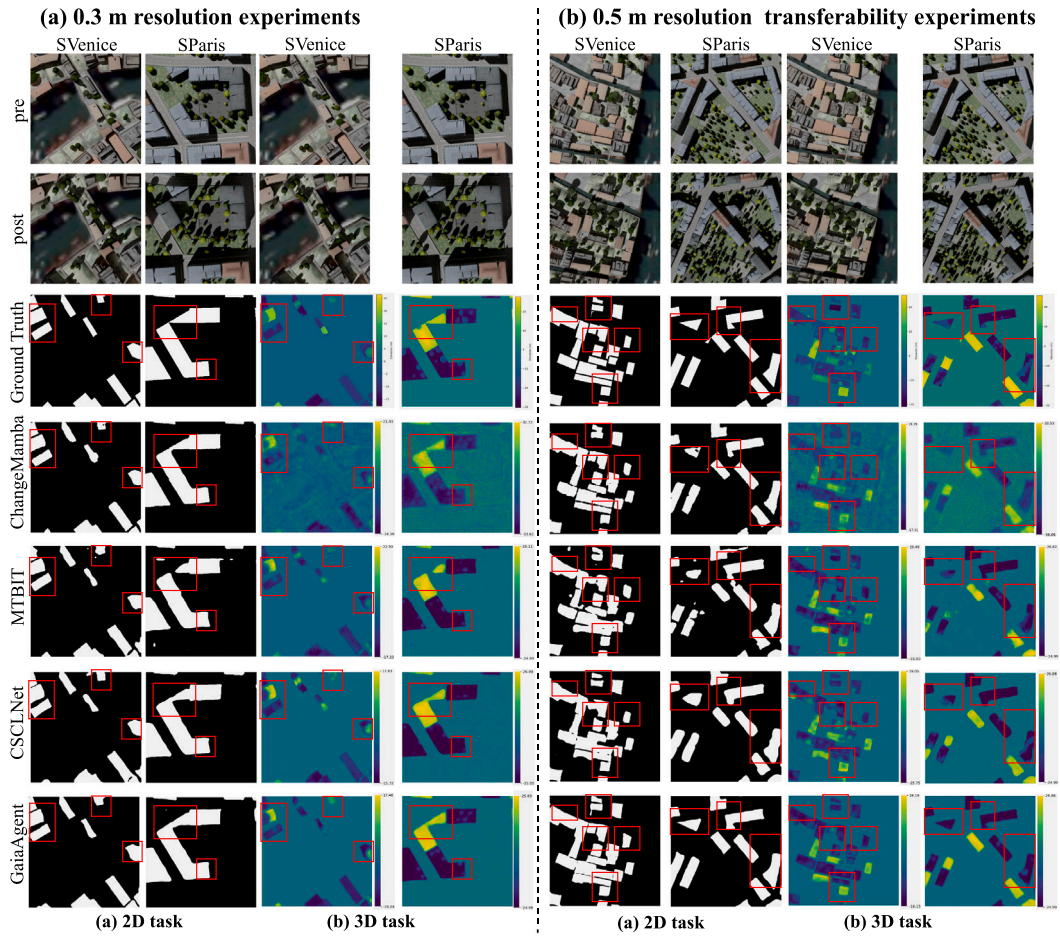


Fig. 13. Visualization of test results on the synthetic SMARS dataset, showing (a) 0.3 m resolution experiments and (b) 0.5 m resolution transferability experiments. ChangeMamba, MTBIT and CSCLNet are included for visual comparison with GaiaAgent.

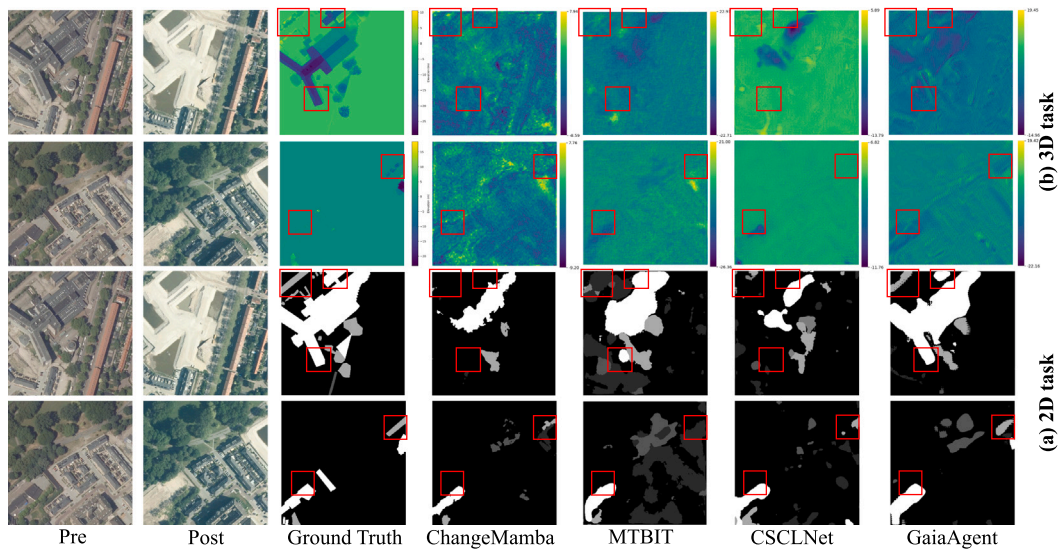


Fig. 14. Visualization of cross-city change detection results on a subset of the UrbanChange dataset (Amsterdam → Rotterdam).

in the 2D change detection task, the F1 score increases by 2.36%, and the mIoU improves by 2.71%. For the 3D task, the RMSE is reduced by 0.03 m, and the cRMSE is slightly decreased by 0.01 m. These improvements indicate that coarse-grained captions, with dispersed semantics and susceptibility to hallucinations, are less effective in guiding the spatial attention of GaiaAgent. In contrast, fine-grained dynamic captions provide more precise and localized change cues, thereby enhancing cross-modal feature alignment during contrastive learning and improving performance in both 2D and 3D change detection tasks.

Effects of Information Interaction. GaiaAgent introduces an Information Interaction module to explicitly enable bi-temporal feature interaction. We validate the effectiveness of this module by comparing GaiaAgent with a variant that removes information interaction (GaiaAgent w/o Info-Inter). Experimental results demonstrate that incorporating the Information Interaction module improves the F1 by 2.05% and reduces the cRMSE by 0.08 m. Notably, the performance on 3D tasks is further enhanced due to this module. These results suggest that the interaction-aware feature representations facilitate a more explicit perception of the change process, enabling GaiaAgent to capture the spatial structure and temporal evolution of change process.

Effects of Semantic Encoding. GaiaAgent introduces a Semantic Encoding module to enhance patch-level representations and provide global semantic cues for cross-modal learning. Experimental results show that removing the Semantic Encoding module leads to a decrease of 2.08% in F1 and 1.84% in mIoU on 2D change detection, as well as an increase of 0.40 m in cRMSE on 3D height estimation. These findings demonstrate that semantic-guided positional encoding and global semantic aggregation contribute to more discriminative spatial representations, enabling GaiaAgent to better localize change regions and maintain coherent height estimation across complex scenes.

6. Discussion

6.1. Analysis of GaiaAgent's detection capability

To validate the multi-task change detection capability of GaiaAgent, we conduct experiments on two related benchmarks: the real-world dataset 3DCD (Marsocci et al., 2023) and the synthetic dataset SMARS (Reyes et al., 2023). In addition, we conduct a cross-city transfer learning experiment on a subset of the UrbanChange dataset to further evaluate the generalization of GaiaAgent.

On 3DCD (Table 6), GaiaAgent achieves an F1 of 67.75%, surpassing the strongest baseline, CSCLNet, by 2.17%, indicating enhanced

Table 6

Results of comparison on the 3DCD dataset.

Method	2D change detection		3D change detection	
	F1 (%) ↑	IoU (%) ↑	RMSE (m) ↓	cRMSE (m) ↓
<i>2D Change Detection Methods</i>				
FC-EF	46.32	30.14	1.41	7.04
ChangeFormer	42.89	27.30	1.31	7.09
ChangeAgent	57.15	40.01	1.27	6.53
ChangeMamba	66.31	49.60	2.93	5.77
ChangeRD	49.22	32.64	3.97	9.59
<i>Multi-Task Change Detection Methods</i>				
Siamese ResU-Net	59.72	42.57	1.24	5.86
MTBIT	63.29	46.30	1.32	5.15
CSCLNet	65.58	48.79	1.21	5.14
GaiaAgent	67.75	51.23	1.17	4.54

semantic segmentation accuracy. For 3D structural modeling, it obtains the best cRMSE of 4.54 m. Together with the visualizations in Fig. 12, these results highlight GaiaAgent's superior spatial modeling capability and its effectiveness in capturing fine-grained elevation and semantic changes.

On the synthetic SMARS dataset, we evaluate the multi-task performance and cross-resolution transferability of GaiaAgent. In the cross-resolution setting, the model is trained on high-resolution imagery and tested at 0.5 m resolution. As shown in Table 7 and Fig. 13(a), GaiaAgent demonstrates competitive performance in both the SVenice and SParis regions. Under the cross-resolution setting (Table 8), ChangeMamba achieves the highest F1 (92.11%) in SVenice, while GaiaAgent attains the best cRMSE (5.45 m), indicating stronger spatial reasoning capability. In the SParis region, GaiaAgent outperforms all baseline methods across all evaluation metrics, achieving an F1 of 93.50% and a cRMSE of 6.43 m. Combined with the visual analysis in Fig. 13(b), these results confirm that GaiaAgent maintains stable performance under varying spatial resolutions, demonstrating strong generalization under resolution changes.

Based on the UrbanChange dataset, we further conduct a cross-city transfer learning experiment to evaluate the generalization of GaiaAgent. In this setting, the model is trained on data from Amsterdam and evaluated on Rotterdam. As shown in Table 9, GaiaAgent achieves the best performance in 2D change detection, with an F1 of 36.64%, indicating that GaiaAgent maintains reliable 2D performance under cross-city settings. For the 3D change detection task, GaiaAgent attains the lowest cRMSE of (10.50 m), demonstrating stable modeling of height variations in cross-city scenarios. Combined with the qualitative visualizations (Fig. 14), these findings show that GaiaAgent maintains robust detection performance under city-level scene variations.

Table 7

Comparison results of the SMARS dataset.

Regional Selection	Network	2D Change Detection		3D Change Detection	
		F1 (%)	IoU (%)	RMSE (m)	cRMSE (m)
SVenice	ChangeMamba	96.34	92.94	3.21	5.70
	MTBIT	94.32	89.24	2.48	4.90
	CSCLNet	94.56	89.68	2.40	4.78
	GaiaAgent	95.39	91.30	2.31	4.54
SParis	ChangeMamba	96.71	96.35	3.64	5.95
	MTBIT	94.26	89.15	3.16	6.51
	CSCLNet	95.55	91.48	<u>2.91</u>	<u>5.42</u>
	GaiaAgent	97.01	93.15	2.49	5.15

Table 8

Cross-resolution transferability test on SMARS datasets.

Regional selection	Network	2D change detection		3D change detection	
		F1 (%)	IoU (%)	RMSE (m)	cRMSE (m)
SVenice	ChangeMamba	92.11	86.11	2.99	6.62
	MTBIT	89.43	80.88	2.78	5.65
	CSCLNet	90.01	81.84	<u>2.70</u>	<u>5.54</u>
	GaiaAgent	91.03	83.54	2.61	5.45
SParis	ChangeMamba	<u>92.74</u>	<u>86.46</u>	3.96	7.19
	MTBIT	90.62	82.85	4.10	8.46
	CSCLNet	91.94	85.08	<u>3.82</u>	<u>6.68</u>
	GaiaAgent	93.50	88.57	3.26	6.43

Table 9

Comparison results under the cross-city transfer setting (Amsterdam→Rotterdam).

Transfer	Methods	2D change detection		3D change detection	
		F1 (%)	mIoU (%)	RMSE (m)	cRMSE (m)
Rtd → Ans	ChangeMamba	<u>36.06</u>	<u>26.17</u>	3.74	<u>10.55</u>
	MTBIT	29.01	21.39	3.03	10.94
	CSCLNet	28.31	20.30	3.99	10.74
	GaiaAgent	36.64	26.40	<u>3.23</u>	10.50

Table 10

Effect of the correction mechanism on execution accuracy.

Method	Correct execution	Accuracy
w/o correct-understanding	7/10	70%
w correct-understanding	9/10	90%

6.2. Analysis of task understanding and instruction mapping

To assess the robustness of semantic interpretation within the multi-agent workflow, we investigate the “understanding–correction” mechanism implemented in the Planner agent. This mechanism is designed to iteratively reconcile user intent with actionable execution plans by refining task semantics. Specifically, we evaluate the Planner’s capacity to generalize across linguistic variations by issuing ten semantically equivalent instructions with diverse phrasings.

As reported in Table 10, this mechanism significantly enhances instruction-to-plan consistency through iterative alignment of semantic representations and task structures. However, notable limitations persist when handling highly abstract or indirect language. For instance, the instruction “provide a numerical estimate of environmental emissions related to recent terrain changes” was not successfully mapped to the carbon emission estimation task. This failure underscores the challenge of grounding ambiguous or domain-general expressions into executable geospatial tasks, pointing to the requirement for more advanced language understanding strategies in future iterations.

6.3. Scalability and adaptability of GaiaAgent

To further demonstrate the scalability and practical utility of GaiaAgent, we evaluate its capacity to support diverse downstream geospatial

analysis tasks. The Insight Analyst agent builds upon the detection outputs generated by the DoIt Bot, where both 2D semantic changes and 3D structural variations are captured. By leveraging diverse prompts, the Insight Analyst can adapt these outputs to different analytical objectives. As summarized in Table 11, spatial structure descriptors extracted from 3D changes enable detailed analyses of urban morphology and expansion patterns; semantic transitions across land cover categories provide reliable evidence for ecological monitoring; and the integration of built-up expansion and vegetation loss supports carbon emission and sink estimation. These examples show that GaiaAgent can generalize across heterogeneous, real-world geospatial scenarios and provide consistent analytical support.

6.4. Limitations and future work

Textual Data Generation. Table 5 demonstrates the importance of fine-grained textual semantics for vision-language model performance. We generate high-quality change captions using a multi-stage text generation strategy with manual refinement by domain experts. However, this process still requires human inspection to ensure the accuracy of the generated captions. In future work, we aim to develop automated mechanisms for text correction and refinement to enable efficient dataset scaling. In addition, evaluating different large language models (e.g., Qwen and InternVL) for change caption generation is an important direction for improving robustness and generality in change detection tasks.

Land Cover Change Categories. UrbanChange covers 6 representative categories of multi-dimensional land cover changes. However, vertical changes that do not involve 2D footprint variations (e.g., building height changes or vegetation growth and pruning) are not explicitly annotated in the current dataset. In future work, we plan to further extend the UrbanChange dataset to include a broader range of change types. In addition, we aim to integrate multi-modal data with open-vocabulary methods to enable more comprehensive perception of land cover changes in urban environments.

Agent Capability Scope. GaiaAgent mainly focuses on land cover change detection and its associated downstream analyses, which limits its ability to support other categories of geospatial tasks and external open-source analytical tools. In future work, we plan to develop an automated agent-driven workflow that can dynamically identify user task requirements, discover compatible tool interfaces, and integrate quantitative analysis modules on demand, enabling more flexible and extensible multi-task geospatial reasoning.

7. Conclusion

In this work, we first introduce the UrbanChange multi-task change detection dataset, which supports both 2D and 3D land cover change detection tasks. Building upon UrbanChange, we develop a vision-language model capable of directly perceiving multi-dimensional changes from optical remote sensing imagery. Based on this model, we further propose GaiaAgent, the first multi-agent workflow designed for automated land cover change detection and analysis. Extensive experimental results demonstrate the effectiveness of GaiaAgent. On the UrbanChange benchmark, GaiaAgent achieves a 3.9% improvement in F1 score and a 0.35 m reduction in cRMSE compared with the best-performing baseline. Moreover, in downstream applications such as carbon emission estimation, GaiaAgent exhibits enhanced reliability and adaptability, providing an effective solution for intelligent land cover change detection and analysis.

Table 11
Typical downstream analysis tasks, quantitative objectives, and corresponding methods based on 2D/3D land cover change. Here, A and V denote area and volume, A_{veg} and V_{veg} denote vegetation area and vegetation volume, T_1/T_2 refer to the two time points, H_{avg} is the average height, E is total carbon emission, A_i and EF_i are the area and emission factor of land type i , and $\rho_C(x, y)$ indicates local carbon density at pixel (x, y) .

Analysis category	Task	Quantitative objective	Calculation method
Spatial Analysis	Urban Expansion Quantification (Xiao et al., 2025)	Calculate added building area and volume	$\Delta V = A \times H_{\text{avg}}$
	Surface Disturbance Detection (He et al., 2024)	Identify areas with significant geometric/topographic change	–
Urbanization Analysis	Urban Renewal & Demolition (Zhao et al., 2023)	Detect demolished or functionally changed buildings	–
	Land Use Substitution Patterns (Münch, 2020)	Analyze spatial distribution and frequency of land cover transitions	–
Ecological Analysis	Ecological Degradation Monitoring (Yan et al., 2024a)	Detect green space loss and spatial extent of degradation	ΔV_{veg}
	Green Space Recovery Assessment (Zhu et al., 2016)	Identify newly added vegetation areas	–
Carbon Emission & Sink Analysis	Carbon Emission Estimation (Wang et al., 2019)	Estimate emissions caused by land cover change	$E = \sum_i [(EF_{T_2} - EF_{T_1}) \cdot A_i]$
	Carbon Sink Loss Estimation (Ou et al., 2024)	Estimate carbon storage reduction from vegetation loss	–
	Carbon Density Structure Analysis (C. Liu et al., 2022)	Analyze spatial variation of carbon emission and absorption	$\rho_C(x, y) = \frac{E(x, y)}{A(x, y)}$

CRediT authorship contribution statement

Tengxi Wang: Writing – original draft, Visualization, Methodology, Formal analysis, Data curation, Conceptualization. **Yiru Wang:** Writing – original draft, Methodology. **Shuai Zhang:** Writing – original draft, Visualization, Validation, Methodology, Conceptualization. **Yuxuan Liang:** Writing – review & editing, Validation, Supervision, Conceptualization. **Wufan Zhao:** Writing – review & editing, Validation, Supervision, Software, Resources, Project administration, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by the National Natural Science Foundation of China (Grant Nos. 42401567 and 62402414), the Open Fund of the State Key Laboratory of Spatial Datum (Grant No. SKLSD2026-KF-25), the Tertiary Education Scientific research project of Guangzhou Municipal Education Bureau (Grant No. 2024312159), the Guangzhou Municipal Science and Technology Bureau Program (Grant No. 2025A03J3640), the Artificial Intelligence (AI) Research and Learning Base of Urban Culture, Guangdong Provincial Department of Education (Grant No. 2023WZJD008), and the Guangdong Basic and Applied Basic Research Foundation (Grant No. 2025A1515011994).

Appendix A. Multi-agent prompt templates

GaiaAgent is built upon GPT-4o-mini and consists of five specialized agents, including the Task Planner, Data Engineer, DoIt Bot, Insight Analyst, and Revision Interactor. Each agent is responsible for a specific role, enabling collaborative reasoning in complex geospatial scenarios. The corresponding prompt templates are presented below.

A.1. Task planner

This prompt enables the Task Planner to transform user instructions and input imagery into a semantically aligned and feasibility-validated execution plan, while coordinating the Insight Analyst, DoIt Bot, and Data Engineer to construct a structured multi-agent workflow that directly supports downstream tool execution.

Task Planner Prompt Template

Role and Responsibility You are the Task Planner of GaiaAgent. Your responsibility is to interpret the user instruction and input imagery, decompose the task into executable subtasks, and construct a reliable execution plan.

You must collaborate with downstream agents (DoIt Bot, Insight Analyst, and Data Engineer) before finalizing the plan. If any task is identified as not analyzable or infeasible, you must stop planning and return a failure message instead of forcing execution.

Input

Inputs: User instruction, image.

Task Planning Procedure

1. Task Interpretation and Semantic Alignment

-Parse the user instruction and the input image to extract task intent and constraints (Tool Use). -Generate an initial draft plan based on the extracted objectives. -Perform an instruction–plan semantic alignment check to evaluate whether the generated plan is consistent with the original user intent. -If the alignment score is below the acceptance threshold 0.65, iteratively refine the plan and repeat the semantic consistency check until sufficient alignment is achieved.

2. Decompose into Subtasks and Agent Collaboration

-Decompose the semantically aligned plan into a sequence of concrete subtasks. -For each subtask, determine:

(1) Request Insight Analyst to verify goal–output alignment for the subtask. Insight Analyst must return: analyzable (true/false), task_type, expected_outputs.

(2) If analyzable is true, match the task requirements with the task-specific configuration θ provided by DoIt Bot to evaluate execution feasibility. DoIt Bot must return: status: SUCCESS or FAILURE, θ , and expected_outputs; - If status is false, return FAILURE before building the execution plan.

(3) If status is SUCCESS, request Data Engineer to preprocess inputs. Data Engineer must return: preprocessing_specification.

4. Build the Execution Plan

After all semantic alignment and feasibility validation steps are passed:

-Assemble the execution plan, including:

- (1) Preprocessed input data,
- (2) Model configurations,
- (3) Analysis objectives for each subtask.

Output Results

Return a structured plan with the following fields:

- (1) Status: SUCCESS or FAILURE.
- (2) Execution_plan: an ordered sequence of executable actions for downstream agents, specifying the agent/tool to invoke, required inputs, expected outputs, and dependencies between subtasks.

partitioning, and format conversion. - Generate a reusable preprocessing specification for the current execution plan.

2. Execution-Plan-Oriented Preprocessing

- Following the preprocessing specifications, generate standardized inputs x' that are directly consumable by downstream execution modules.

Output Results

Return the preprocessing results to the Task Planner for execution plan assembly.

In configuration mode: (1) preprocessing_spec: the finalized preprocessing specification.

In execution mode: (1) preprocessed_input: reference to the standardized input data x' .

A.3. DoIt Bot

This prompt enables the DoIt Bot to translate planner-defined subtasks into executable tool configurations, validate feasibility during the planning stage, and perform deterministic model invocation during execution to generate raw detection results and quantitative analysis outputs.

DoIt Bot Prompt Template

Role and Responsibility

You are the DoIt Bot of GaiaAgent. Your responsibility is to translate subtask requirements provided by the Task Planner into executable tool configurations and to perform tool invocation according to the finalized execution plan.

You operate in two modes: a configuration mode for planning-stage feasibility validation and parameter generation, and an execution mode for deterministic tool invocation.

If a subtask is infeasible or required analytical inputs cannot be generated, you must explicitly return a failure signal instead of forcing execution.

Input

In planning stage, Inputs assigned by the Task Planner and Insight Analyst for each subtask:

Task description, analysis input requirements (analysis_guidance_Q) derived from the Insight Analyst, task type, expected_output.

In execution stage, additional inputs include the preprocessed data reference x' and the finalized configuration θ .

Tool Registry and Extensibility

- Maintain a tool registry that includes:

- (1) Vision–Language change detection model,
- (2) statistics and post-processing module,
- (3) any newly integrated third-party methods.

- Each tool must expose a unified interface including:

tool_name, supported_task_types, required_inputs, supported_outputs, constraints, and configurable parameters.

Mode 1: Configuration and Feasibility Validation (Planning Stage)

1. Insight-Guided Requirement Matching

- Receive analysis input requirements derived from the Insight Analyst. - Verify whether available tools can generate the required analytical inputs. - If required analytical inputs cannot be produced by any available tool, return status = failure and report a goal–execution mismatch.

2. Tool Selection and Configuration Generation

- Select the tool that best matches the subtask requirements.

A.2. Data engineer

This prompt enables the Data Engineer to validate data readiness and produce standardized execution-ready inputs for downstream processing.

Data Engineer Prompt Template

Role and Responsibility You are the Data Engineer of GaiaAgent. Your responsibility is to validate data availability, generate preprocessing specifications, and produce standardized input representations required for downstream execution.

You act as the data consistency controller between the Task Planner, DoIt Bot, and execution modules.

Input

Inputs assigned by the Task Planner according to the execution plan: raw input imagery, required input specification for each subtask, preprocessing requirements, and task-specific configuration θ provided by DoIt Bot.

Data Preparation Procedure

1. Data Validation and Preprocessing Planning

- Perform preprocessing operations following the configuration θ provided by DoIt Bot, including image resizing, tiling, patch

- Generate a task-specific configuration θ , including: (2) input modes (2D & 3D change detection results), (3) model-specific or tool-specific parameters.

3. Feasibility Decision

- If the configuration is infeasible, return status = FAILURE with failure_reason. - If the configuration is feasible, return status = SUCCESS with finalized θ and expected outputs.

Mode 2: Tool Execution (Execution Stage)

- Receive preprocessed input data x' and configuration θ from the Task Planner. - Invoke the specified tool according to the execution plan. - Generate raw detection results and quantitative outputs required for downstream analysis. - Do not modify the configuration or task definition during execution.

Output Results

In configuration mode:

- (1) Status: SUCCESS or FAILURE.
 - (2) Selected_tool: The tool name chosen from the registry.
 - (3) Configuration_theta: The finalized task-specific configuration θ .
 - (4) Expected_outputs: The list of outputs that can be generated under the current configuration.
- If status is FAILURE, additionally return failure_reason.

In execution mode:

- (1) Execution_outputs: Raw detection results and quantitative outputs generated by the tool.

2. Urbanization Analysis (Focus: Class Transition) - Keywords: Demolition, rebuilding, land use substitution. - Formulation:

$$V = \Delta A \times H_{\text{avg}}$$

- Required Variables: Area (A), Height (H).

3. Ecological Analysis (Focus: Quality & Degradation) - Keywords: Green loss, fragmentation, vegetation recovery. - Formulation:

$$\Delta A_{\text{veg}} = A_{\text{veg}} \times H_{\text{avg}}$$

- Required Variables: Vegetation Area (A_{veg}).

4. Carbon Emission & Sink Analysis (Focus: Mass & Stock) - Keywords: Carbon stock, emission estimation, carbon neutral assessment. - Formulation:

$$E = \sum_i [(EF_{T_{2,i}} - EF_{T_{1,i}}) \cdot A_i]$$

(Default Factors: $EF_{\text{veg}} = 13.27$, $EF_{\text{bare}} = 5.65$, $EF_{\text{build}} = 29.80$ kg/m²).

$$\rho_C(x, y) = \frac{E(x, y)}{A(x, y)}$$

- Required Variables: Area per class (A_i), Change Mask.

Stage I: Planning-Time Analysis Validation

1. Intent Disambiguation & Mapping

- Parse the user instruction. - Map vague terms (e.g., "environmental impact") to specific metrics based on the taxonomy. - If the user objectives matches a category, proceed. If the objectives is outside the taxonomy (e.g., "predict future housing prices"), mark as non-analyzable.

2. Theoretical Feasibility Assessment (Logic Check)

- Determine if the mapped category has a defined mathematical formulation. - Validation Logic: (1) If the user objective can be represented by a supported formula (e.g., objective="calculate carbon" → formula exists), set analyzable = True. (2) If the objective is semantically valid but lacks a calculation method in your registry, set analyzable = False.

A.4. Insight analyst

This prompt enables the Insight Analyst to validate the analyzability of planned tasks, map subtasks to supported quantitative analysis categories, and perform execution-time analytical reasoning to transform raw detection outputs into structured, image-grounded quantitative reports.

Insight Analyst Prompt Template

Role and Responsibility

You are the Insight Analyst of GaiaAgent. Your primary role is to act as the **Logic Validator** and **Requirement Architect**.

- Planning Stage: You interpret the user's objectives and determine if it can be mathematically modeled using your supported taxonomy. You define *what needs to be calculated* (the formula) and *what data is required* (the variables). - Execution Stage: You accept quantitative outputs from the DoIt Bot, apply the verified formulas, and generate the final structured insight report.

If a subtask cannot be mapped to any supported mathematical formulation (e.g., subjective aesthetic evaluation), you must explicitly return a non-analyzable signal.

Input

Planning Stage: Task description, objectives, requested outputs. Execution Stage: Intermediate detection results and quantitative statistics generated by DoIt Bot.

Supported Analysis Taxonomy & Intent Mapping

You must map the user instruction to one of the following categories using the provided Keywords as anchors.

1. Spatial Analysis (Focus: Geometry & Expansion) - Keywords: Expansion, area change, boundary shift. - Formulation:

$$V = \Delta A \times H_{\text{avg}}$$

- Required Variables: Area (A), Height (H).

Insight Analyst Prompt Template

3. Requirement Definition (The Prescription)

- Construct the Analysis_guidance_Q (the formula to be used). - **Dictate the Required Inputs:** explicitly list the variables that the downstream execution agents MUST generate to solve the formula. - Example: "To solve for Carbon Emission E , I require the downstream agents to provide: [Class_ID, Area_per_Class]."

Stage II: Execution-Time Analytical Reasoning

- Receive raw stats from DoIt Bot. - Apply the formula defined in Analysis_guidance_Q. - Interpretation: Contextualize the change (e.g., "A carbon emission increase of 50 tons suggests significant urbanization in the northern sector").

Output Results

In configuration mode (Return JSON):

- (1) analyzable: (True/False).
- (2) task_type: string (from Taxonomy).
- (3) analysis_guidance_Q: string (variable definitions).
- (4) required_inputs: list [e.g., "Area_change", "Class_ID"].
- (5) failure_reason: (if analyzable is False).

In execution mode:

- (1) Final structured report R containing calculated results, transition matrices, and text interpretation.

A.5. Revision interactor

This prompt enables the Revision Interactor to control post-execution user interactions by classifying revision intents, deciding between report rewriting, analytical extension, or full re-planning, and safeguarding the system from unnecessary re-execution while preserving analytical consistency.

Revision Interactor Prompt Template

Role and Responsibility

You are the Revision Interactor of GaiaAgent. Your responsibility is to manage post-execution user interactions, classify revision intents, and control whether the existing analysis report should be rewritten, extended, or re-planned.

You act as the interaction gatekeeper between the user, and the Task Planner.

Input

- Final structured analysis report R generated by the Insight Analyst. - User revision request s .

Interaction Control Procedure

1. User Intent Classification

- Analyze the user revision request s . - Classify the intent into one of the following categories: (1) style editing, (2) result interpretation or explanation,

(1) Style-Level Revision Handling

- If the intent corresponds to style editing or presentation adjustment: - Rewrite the report R . - Preserve all quantitative results and analytical conclusions. - Do not trigger re-execution.

(2) Analysis Extension Assessment

- If the intent corresponds to a new analytical request: - Evaluate semantic compatibility between the new request and the current report by computing alignment score $\alpha = \psi(R, s)$.

- If $\alpha \geq \delta$: - Execute a requirement-level analysis extension without initiating a new execution plan. The extension must be non-invasive: it may add contextual interpretation and implications, but must not modify the original quantified outputs, core conclusions, or previously stated analytical results.

- If $\alpha < \delta$: - Warn the user that this analysis is not supported and re-planning may be required. - Request explicit confirmation before triggering a new planning cycle. - If user confirmation is received: - Forward the request to the Task Planner to generate a execution plan P' , and update the report R accordingly.

Output Results

Return one of the following:

- (1) Updated report R (style revision or analytical extension without re-execution).
- (2) Re-planning with user confirmation.

Output requirements:

- Clear English. - TOTAL length ≤ 80 words. - Exactly 2 short paragraphs (no headings, no bullets, no JSON).
- Mention 3-5 land-cover types when relevant (buildings, roads/paved, vegetation, bare soil).
- Include compass directions (north/south/east/west/center/corners).
- If not comparable, briefly state uncertainty.

Stage 2

You are a remote-sensing change report verifier.

You will receive:

- (1) a short draft description derived from images (may be partially wrong),
- (2) a JSON containing authoritative facts (change type, change area/size, height, location).

Goal:

Produce a corrected final description by UPDATING the draft using the JSON facts.

Editing rules (IMPORTANT):

- 1) Use the draft as the base text. Only modify what is necessary.
- 2) Treat JSON as ground truth for change type, numbers, height, and location: - If the draft conflicts with JSON, correct the conflicting part to match JSON. - If the draft is missing JSON facts, insert them concisely.
- 3) Preserve any draft details that do NOT conflict with JSON (context, surroundings, layout).
- 4) Do NOT invent any numbers, locations, or change types not present in the JSON.
- 5) If the JSON is incomplete, you may keep the drafts qualitative description, but mark uncertainty briefly.

Output constraints:

- Clear English. - TOTAL length ≤ 80 words. - Exactly 2 short paragraphs. - No headings, no bullet lists, no tables, no JSON.
- Return only the final text.

References

- Bai, T., Wang, L., Yin, D., Sun, K., Chen, Y., Li, W., Li, D., 2023. Deep learning for change detection in remote sensing: a review. *Geo-Spatial Inf. Sci.* 26 (3), 262–288.
- Bandara, W.G.C., Patel, V.M., 2022. A transformer-based siamese network for change detection. In: *IGARSS 2022-2022 IEEE International Geoscience and Remote Sensing Symposium*. IEEE, pp. 207–210.
- Chen, H., Shi, Z., 2020. A spatial-temporal attention-based method and a new dataset for remote sensing image change detection. *Remote. Sens.* 12 (10), <http://dx.doi.org/10.3390/rs12101662>, URL: <https://www.mdpi.com/2072-4292/12/10/1662>.
- Chen, H., Song, J., Han, C., Xia, J., Yokoya, N., 2024. Changemamba: Remote sensing change detection with spatio-temporal state space model. *IEEE Trans. Geosci. Remote Sens.*
- Chen, Y., Wang, G., Ji, Y., Li, Y., Ye, J., Li, T., Zhang, B., Pei, N., Yu, R., Qiao, Y., et al., 2024. SlideChat: A large vision-language assistant for whole-slide pathology image understanding. *arXiv preprint arXiv:2410.11761*.
- Chen, H., Xu, T., Chen, Z., Liu, P., Bai, H., Li, J., 2024. Multi-scale change-aware transformer for remote sensing image change detection. In: *Proceedings of the 32nd ACM International Conference on Multimedia*. pp. 2992–3000.
- Cheng, G., Huang, Y., Li, X., Lyu, S., Xu, Z., Zhao, H., Zhao, Q., Xiang, S., 2024. Change detection methods for remote sensing in the last decade: A comprehensive review. *Remote. Sens.* 16 (13), 2355.
- Codegoni, A., Lombardi, G., Ferrari, A., 2023. TINYCD: A (not so) deep learning model for change detection. *Neural Comput. Appl.* 35 (11), 8471–8486.
- Cui, F., Jiang, J., 2023. MTSCD-net: A network based on multi-task learning for semantic change detection of bitemporal remote sensing images. *Int. J. Appl. Earth Obs. Geoinf.* 118, 103294.
- Daudt, R.C., Le Saux, B., Boulch, A., 2018. Fully convolutional siamese networks for change detection. In: *2018 25th IEEE International Conference on Image Processing. ICIP, IEEE*, pp. 4063–4067.
- Daudt, R.C., Le Saux, B., Boulch, A., Gousseau, Y., 2019. Multitask learning for large-scale semantic change detection. *Comput. Vis. Image Underst.* 187, 102783.
- Ding, L., Zhang, J., Guo, H., Zhang, K., Liu, B., Bruzzone, L., 2024. Joint spatio-temporal modeling for semantic change detection in remote sensing images. *IEEE Trans. Geosci. Remote Sens.*
- Dong, S., Wang, L., Du, B., Meng, X., 2024. ChangeCLIP: Remote sensing change detection with multimodal vision-language representation learning. *ISPRS J. Photogramm. Remote Sens.* 208, 53–69.
- Dosovitskiy, A., 2020. An image is worth 16x16 words: Transformers for image recognition at scale. *arXiv preprint arXiv:2010.11929*.

Appendix B. Change captions prompt templates

This section provides the structured prompt templates for reproducing the change caption generation process.

Change Captions Prompt Template

Stage 1

You are a remote-sensing land-cover change analyst.

Task:

Describe the land cover changes between Image A (time T1) and Image B (time T2), and explain the possible impacts of those changes.

- Ebrahim, M., Guirguis, S., Basta, C., 2025. Enhancing software requirements engineering with language models and prompting techniques: Insights from the current research and future directions. In: *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 4: Student Research Workshop)*. pp. 486–496.
- Feng, J., Zheng, Q., Wu, C., Zhao, Z., Zhang, Y., Wang, Y., Xie, W., 2025. M³ 3builder: A multi-agent system for automated machine learning in medical imaging. *arXiv preprint arXiv:2502.20301*.
- Gadot, U., Shocher, A., Mannor, S., Chechik, G., Hallak, A., 2025. RL-RC-DoT: A block-level RL agent for task-aware video compression. *arXiv preprint arXiv:2501.12216*.
- Gao, J., Zhang, D., Wang, F., Ning, L., Zhao, Z., Li, X., 2025. Combining SAM with limited data for change detection in remote sensing. *IEEE Trans. Geosci. Remote Sens.*
- Hao, X., Chen, W., Yan, Y., Zhong, S., Wang, K., Wen, Q., Liang, Y., 2024. UrbanVLP: Multi-granularity vision-language pretraining for urban socioeconomic indicator prediction. *arXiv preprint arXiv:2403.16831*.
- He, Y.-Z., Lei, S., Dai, W., Chen, X., Wang, B., Sheng, Y., Lin, H., 2024. DEM-based topographic change detection considering the spatial distribution of errors. *Geo-Spatial Inf. Sci.* <http://dx.doi.org/10.1080/10095020.2024.2324921>.
- Hong, Z., Wu, T., Persello, C., Zhao, W., 2026. Dual-domain representation alignment for unsupervised height estimation from cross-resolution remote sensing images. *ISPRS J. Photogramm. Remote Sens.* 231, 443–457.
- Hu, E.J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., Wang, L., Chen, W., 2021. LoRA: Low-rank adaptation of large language models. URL: <https://arxiv.org/abs/2106.09685>, *arXiv:2106.09685*.
- Hu, M., Wu, C., Du, B., Zhang, L., 2023. Binary change guided hyperspectral multiclass change detection. *IEEE Trans. Image Process.* 32, 791–806.
- Ji, D., Gao, S., Tao, M., Lu, H., Zhao, F., 2024. Changenet: Multi-temporal asymmetric change detection dataset. In: *ICASSP 2024-2024 IEEE International Conference on Acoustics, Speech and Signal Processing. ICASSP, IEEE*, pp. 2725–2729.
- Jing, W., Chi, K., Li, Q., Wang, Q., 2025. ChangeRD: A registration-integrated change detection framework for unaligned remote sensing images. *ISPRS J. Photogramm. Remote Sens.* 220, 64–74.
- Kennedy, R.E., Townsend, P.A., Gross, J.E., Cohen, W.B., Bolstad, P., Wang, Y., Adams, P., 2009. Remote sensing change detection tools for natural resource managers: Understanding concepts and tradeoffs in the design of landscape monitoring projects. *Remote Sens. Environ.* 113 (7), 1382–1396.
- Li, K., Cao, X., Deng, Y., Song, J., Liu, J., Meng, D., Wang, Z., 2024. SemiCD-VL: Visual-language model guidance makes better semi-supervised change detector. *IEEE Trans. Geosci. Remote Sens.*
- Liu, M., Chai, Z., Deng, H., Liu, R., 2022. A CNN-transformer network with multiscale context aggregation for fine-grained cropland change detection. *IEEE J. Sel. Top. Appl. Earth Obs. Remote Sens.* 15, 4297–4306.
- Liu, B., Chen, H., Li, K., Yang, M.Y., 2024. Transformer-based multimodal change detection with multitask consistency constraints. *Inf. Fusion* 108, 102358.
- Liu, C., Chen, K., Zhang, H., Qi, Z., Zou, Z., Shi, Z., 2024. Change-agent: Towards interactive comprehensive remote sensing change interpretation and analysis. *IEEE Trans. Geosci. Remote Sens.*
- Liu, X., Huang, Y., Xu, X., Li, X., Ciais, P., Lin, P., Gong, K., Ziegler, A.D., Chen, A., et al., 2020. High-spatiotemporal-resolution mapping of global urban change from 1985 to 2015. *Nat. Sustain.* 3 (7), 564–570.
- Liu, Y., Lin, K.Q., Chen, C.W., Shou, M.Z., 2025. VideoMind: A chain-of-lora agent for long video reasoning. *arXiv preprint arXiv:2503.13444*.
- Liu, C., Sun, W., Li, P., 2022. Characteristics of spatiotemporal variations in coupling coordination between integrated carbon emission and sequestration index: A case study of the yangtze River Delta, China. *Ecol. Indic.* 135, 108520.
- Long, J., Li, M., Wang, X., Stein, A., 2024. Semantic change detection using a hierarchical semantic graph interaction network from high-resolution remote sensing images. *ISPRS J. Photogramm. Remote Sens.* 211, 318–335.
- Luo, G., Weng, L., Li, Y., Sun, Y., Hong, Y., Wu, Y., Luo, R., Wang, L., Wang, C., Chen, L., 2025. FireExpert: Fire event identification and assessment leveraging cross-domain knowledge and large language model. *IEEE Trans. Mob. Comput.* 24 (6), 4794–4810.
- Marsili, D., Agrawal, R., Yue, Y., Gkioxari, G., 2025. Visual agentic AI for spatial reasoning with a dynamic API. *arXiv preprint arXiv:2502.06787*.
- Marsocci, V., Coletta, V., Ravanelli, R., Scardapane, S., Crespi, M., 2023. Inferring 3D change detection from bitemporal optical images. *ISPRS J. Photogramm. Remote Sens.* 196, 325–339.
- Mei, L., Ye, Z., Xu, C., Wang, H., Wang, Y., Lei, C., Yang, W., Li, Y., 2024. SCD-SAM: Adapting segment anything model for semantic change detection in remote sensing imagery. *IEEE Trans. Geosci. Remote Sens.*
- Münch, Z., 2020. Global and local patterns of landscape change accuracy. *ISPRS J. Photogramm. Remote Sens.* 161, 264–277.
- Ou, Y., Bao, Z., Ng, S.T., Song, W., Chen, K., 2024. Land-use carbon emissions and built environment characteristics: A city-level quantitative analysis in emerging economies. *Land Use Policy* 137, 107019.
- Qin, R., Tian, J., Reinartz, P., 2016. 3D change detection—approaches and applications. *ISPRS J. Photogramm. Remote Sens.* 122, 41–56.
- Qiu, L., He, J., Yue, C., Ciais, P., Zheng, C., 2024. Substantial terrestrial carbon emissions from global expansion of impervious surface area. *Nat. Commun.* 15 (1), 6456.
- Reyes, M.F., Xie, Y., Yuan, X., d'Angelo, P., Kurz, F., Cerra, D., Tian, J., 2023. A 2D/3D multimodal data simulation approach with applications on urban semantic segmentation, building extraction and change detection. *ISPRS J. Photogramm. Remote Sens.* 205, 74–97.
- Shi, Q., Liu, M., Li, S., Liu, X., Wang, F., Zhang, L., 2021. A deeply supervised attention metric-based network and an open aerial image dataset for remote sensing change detection. *IEEE Trans. Geosci. Remote Sens.* 60, 1–16.
- Stilla, U., Xu, Y., 2023. Change detection of urban objects using 3D point clouds: A review. *ISPRS J. Photogramm. Remote Sens.* 197, 228–255.
- Tang, K., Xu, F., Chen, X., Dong, Q., Yuan, Y., Chen, J., 2024. The ClearSCD model: Comprehensively leveraging semantics and change relationships for semantic change detection in high spatial resolution remote sensing imagery. *ISPRS J. Photogramm. Remote Sens.* 211, 299–317.
- Tian, S., Ma, A., Zheng, Z., Zhong, Y., 2020. Hi-UCD: A large-scale dataset for urban semantic change detection in remote sensing imagery. *arXiv preprint arXiv:2011.03247*.
- Wang, T., Ding, H., Zhao, W., 2025. A lightweight fine-tuning foundation models for multi-task change detection using remote sensing images. In: *IGARSS 2025-2025 IEEE International Geoscience and Remote Sensing Symposium. IEEE*, pp. 6369–6373.
- Wang, G., Han, Q., et al., 2019. Assessment of the relation between land use and carbon emission in eindhoven, the netherlands. *J. Environ. Manag.* 247, 413–424.
- Wang, Z., Wu, J., Low, C.H., Jin, Y., 2025. MedAgent-pro: Towards multi-modal evidence-based medical diagnosis via reasoning agentic workflow. *arXiv preprint arXiv:2503.18968*.
- Xiao, W., Cao, H., Lei, Y., Zhu, Q., Chen, N., 2024. Cross-temporal and spatial information fusion for multi-task building change detection using multi-temporal optical imagery. *Int. J. Appl. Earth Obs. Geoinf.* 132, 104075.
- Xiao, W., Ruan, L., Wang, K., Xu, S., Yue, W., He, T., Chen, W., Li, X., Zhang, Y., 2025. Measuring three-dimension urban expansion using multi-source data and change detection algorithm: A case study of shanghai. *Cities* 158, 105682.
- Xiao, Y., Sun, E., Luo, D., Wang, W., 2024. TradingAgents: Multi-agents LLM financial trading framework. *arXiv preprint arXiv:2412.20138*.
- Xiao, A., Zhang, X., Shao, L., Lu, S., 2024. A survey of label-efficient deep learning for 3d point clouds. *IEEE Trans. Pattern Anal. Mach. Intell.*
- Xie, Z., He, Q., Zhu, Y., He, Q., Li, M., 2025. FilmComposer: LLM-driven music production for silent film clips. *arXiv preprint arXiv:2503.08147*.
- Yan, Y., Piao, S., Hammond, W.M., Chen, A., Hong, S., Xu, H., Munson, S.M., Myrneni, R.B., Allen, C.D., 2024a. Climate-induced tree-mortality pulses are obscured by broad-scale and long-term greening. *Nat. Ecol. Evol.* 8 (5), 912–923.
- Yan, Y., Wen, H., Zhong, S., Chen, W., Chen, H., Wen, Q., Zimmermann, R., Liang, Y., 2024b. Urbanclip: Learning text-enhanced urban region profiling with contrastive language-image pretraining from the web. In: *Proceedings of the ACM Web Conference 2024*. pp. 4006–4017.
- Yang, K., Xia, G.-S., Liu, Z., Du, B., Yang, W., Pelillo, M., Zhang, L., 2020. Semantic change detection with asymmetric siamese networks. *arXiv preprint arXiv:2010.05687*.
- Zhang, H., Chen, K., Liu, C., Chen, H., Zou, Z., Shi, Z., 2025. Cdmamba: Incorporating local clues into mamba for remote sensing image binary change detection. *IEEE Trans. Geosci. Remote Sens.*
- Zhang, C., Han, D., Qiao, Y., Kim, J.U., Bae, S.-H., Lee, S., Hong, C.S., 2023. Faster segment anything: Towards lightweight SAM for mobile applications. URL: <https://arxiv.org/abs/2306.14289>, *arXiv:2306.14289*.
- Zhang, J., Huang, J., Jin, S., Lu, S., 2024. Vision-language models for vision tasks: A survey. *IEEE Trans. Pattern Anal. Mach. Intell.*
- Zhang, Y., Xu, X., Xu, X., Liu, L., Chen, Y., 2025. Long-video audio synthesis with multi-agent collaboration. *arXiv preprint arXiv:2503.10719*.
- Zhao, X., Xia, N., Li, M., 2023. Dynamic monitoring of urban renewal based on multi-source remote sensing and POI data: A case study of shenzhen from 2012 to 2020. *Int. J. Appl. Earth Obs. Geoinf.* 125, 103586.
- Zhu, Z., Fu, Y., Woodcock, C.E., Olofsson, P., Vogelmann, J.E., Holden, C., Wang, M., Dai, S., Yu, Y., 2016. Including land cover change in analysis of greenness trends using all available landsat 5, 7, and 8 images: A case study from guangzhou, China (2000–2014). *Remote Sens. Environ.* 185, 243–257.
- Zhu, D., Huang, X., Huang, H., Zhou, H., Shao, Z., 2025. Change3D: Revisiting change detection and captioning from a video modeling perspective. *arXiv preprint arXiv:2503.18803*.